EXPLAINABLE ADVERSARIAL DRIFT DETECTION FOR MLOPS FEATURE MONITORING

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ABSTRACT

Machine learning models in production suffer silent performance degradation from distribution drift. Existing unsupervised detectors signal *that* drift occurred but not *which* features drifted or *what* corrective action to take. This thesis proposes Explainable Adversarial Drift Detection (EADD), integrating discriminative classification with an entropy-based Prediction Uncertainty Index for early warning, adaptive sequential permutation testing, SHAP interaction-aware drift localization, and a novel Drift Severity Index for automated remediation prescriptions. Evaluated against seven state-of-the-art detectors across synthetic and real-world benchmarks, EADD uniquely achieved complete drift-type coverage with near-zero false alarms while providing feature-level root-cause attribution ($P@k = R@k = 1.0$). The adaptive permutation test reduced computational overhead by up to 70% compared to fixed-budget approaches while maintaining statistical rigor.

IMPLICATION OF THE THESIS

KEY WORDS : DRIFT DETECTION / ADVERSARIAL VALIDATION / COVARIATE DRIFT / MLOPS / DISTRIBUTION SHIFT / STREAMING DATA

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CHAPTER I
INTRODUCTION

1.1 Motivation

Machine learning models deployed in production environments operate under the assumption that future data will resemble training data. This assumption is routinely violated: the statistical properties of input features, label distributions, or the conditional relationship between inputs and outputs evolve over time—a phenomenon collectively termed *distribution drift* (Lu et al., 2018; Gama et al., 2014). Left undetected, drift degrades model performance silently, accumulating technical debt and operational risk (Sculley et al., 2015; Bayram et al., 2022). Widmer and Kubat (1996) established that early detection of such changes is essential for maintaining reliable inference in non-stationary environments.

The practical consequences are well documented. Vela et al. (2022) demonstrated that clinical prediction models exhibit measurable accuracy degradation within months of deployment due to temporal distribution shift. Nestor et al. (2019) corroborated this finding, showing that models for common clinical tasks failed to generalise across temporally shifted patient cohorts within the same hospital system. In high-stakes domains, a binary "drift detected" alarm is insufficient; practitioners require immediate root-cause diagnostics to determine whether degradation stems from specific feature pipelines, population-level distributional shifts, or systemic data quality failures (Sculley et al., 2015).

The drift detection landscape has matured considerably. Statistical methods employ hypothesis tests such as the Kolmogorov–Smirnov test or Maximum Mean Discrepancy to compare reference and current distributions (Rabanser et al., 2019; Gretton et al., 2012). Adaptive windowing methods like ADWIN dynamically adjust historical baselines (Bifet and Gavaldà, 2007). Supervised error-based detectors (DDM, EDDM) monitor model error rates but require ground-truth labels (Gama et al., 2004;

Baena-García et al., 2006). Prediction-based methods track model outputs but depend on calibration quality (Sethi and Kantardzic, 2017; Guo et al., 2017). Despite this diversity, fundamental challenges persist: univariate statistical tests suffer from false alarm storms in high-dimensional settings; supervised methods cannot operate when labels are delayed or unavailable; and no existing method provides integrated root-cause diagnostics (Lu et al., 2018; Webb et al., 2016).

The discriminative paradigm—recasting distribution comparison as a classification problem—offers a principled alternative. Lopez-Paz and Oquab (2017) formalised this as the *classifier two-sample test* (C2ST): if an auxiliary classifier can reliably separate reference from current data based solely on features, the distributions differ significantly. Gözüaçık et al. (2019) adapted this principle for streaming drift detection as the Discriminative Drift Detector (D3), which Lukats et al. (2024) identified as the most robust unsupervised detector in their comprehensive benchmark. However, D3 and all existing discriminative implementations share three critical limitations: (1) they provide only binary detection signals without feature-level attribution, (2) they rely on fixed thresholds lacking statistical justification, and (3) they discard the trained discriminative model without extracting diagnostic value from it.

D3 (Gözüaçık et al., 2019) represents the state-of-the-art unsupervised detector but has two critical operational gaps. First, its fixed AUC threshold (0.7) lacks statistical justification, making it vulnerable to false alarms on correlated and autocorrelated data. Second, it follows a "detect-and-discard" pattern—the trained classifier is discarded after computing AUC, providing no information about *which* features drove the detected drift. Without knowing which features drifted, practitioners cannot determine whether to fix a specific data pipeline, retrain the model, or investigate a data source. Panda et al. (2024) proposed feature-interaction aware hypothesis testing but requires pre-specifying which interactions to test and has quadratic complexity. Feng et al. (2024) provides per-feature performance attributions but is supervised (requires ground-truth labels) and operates post-hoc on collected batches, making it inapplicable when labels are delayed. Hinder et al. (2024c) formalised rigorous mathematical definitions of feature-wise drift but did not provide a streaming detection pipeline. Collectively, no existing work systematically evaluates the accuracy of SHAP-based feature attribution for identifying which features cause drift within discriminative detection—nor integrates such attribution with severity quantification in an unsupervised streaming pipeline.

1.2 Objectives

The primary objective of this thesis is to develop **Enhanced Adversarial Drift Detection (EADD)**, a framework that advances the discriminative drift detection paradigm through three methodological contributions—statistically rigorous adaptive permutation testing with early-warning monitoring, interaction-aware feature attribution, and a novel drift severity metric—integrated into a unified streaming pipeline capable of detecting and diagnosing distribution drift. Specifically, this work aims to:

1. Extend the discriminative drift detection approach of D3 (Gözüaçık et al., 2019) with a non-linear classifier (LightGBM; Ke et al., 2017) and **reservoir sampling** (Vitter, 1985) for representative reference windows, broadening detection coverage to gradual and incremental drift patterns where linear classifiers fail.
2. Introduce **prediction entropy monitoring** (H_{pred}), an early-warning signal derived from the Shannon entropy (Shannon, 1948) of the discriminative classifier's output probabilities, capable of flagging subtle distributional instabilities before they reach the statistical significance threshold.
3. Develop an **adaptive sequential permutation test (ASPT)** that calibrates the number of permutations based on accumulated evidence strength, reducing computational overhead by over 75% compared to fixed-budget approaches while maintaining Type I error control at the nominal level (Gandy, 2009).
4. Propose **SHAP interaction-aware drift localization** that extends beyond marginal feature attribution to identify correlated feature drift through SHAP interaction values (Lundberg et al., 2020), addressing the limitation that marginal SHAP values can miss joint distributional shifts in correlated feature subspaces.

5. Introduce the **Drift Severity Index (DSI)**, a novel composite metric that combines discriminative power (AUC) with attribution concentration (normalised SHAP entropy) to produce a continuous, theoretically motivated drift severity score, replacing the heuristic thresholds used in prior work.
6. Systematically benchmark EADD against seven state-of-the-art unsupervised drift detectors on real-world datasets from the Lukats et al. (2024) benchmark, evaluating detection accuracy, false alarm control, and SHAP feature attribution accuracy.

1.3 Expected Contributions

This thesis makes the following three contributions:

1. Inspired by the statistical analysis of the classifier two-sample test (Lopez-Paz and Oquab, 2017) and its extensions (Pandeva et al., 2024), this thesis introduces an **Adaptive Sequential Permutation Test (ASPT)** (Gandy, 2009) with accompanying **prediction entropy monitoring** (H_{pred}). ASPT monitors evidence accumulation and terminates when statistical significance is established or refuted, reducing computational cost by over 75% compared to fixed-budget permutation testing while preserving Type I error guarantees. H_{pred} provides a continuous early-warning signal from the Shannon entropy of the discriminative classifier's output probabilities, flagging subtle shifts before they reach statistical significance (Chapter 3, Steps 2b–3).
2. Building on SHAP feature distribution analysis (Lundberg and Lee, 2017), this thesis proposes **SHAP interaction-aware drift localisation** (Lundberg et al., 2020) and a novel **Drift Severity Index (DSI)**. The localisation component extends per-feature attribution to SHAP interaction values, enabling detection of correlated feature drift that per-feature methods miss. DSI is defined as $\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s}))$, combining detection

confidence with attribution concentration into a continuous, theoretically motivated severity score (Chapter 3, Step 4).

3. A **comprehensive experimental evaluation** benchmarking EADD against seven state-of-the-art unsupervised detectors on 11 real-world benchmark streams and four synthetic drift types, with formal validation of detection coverage, false alarm control, and SHAP attribution accuracy (Chapter 4).

CHAPTER II
LITERATURE REVIEW

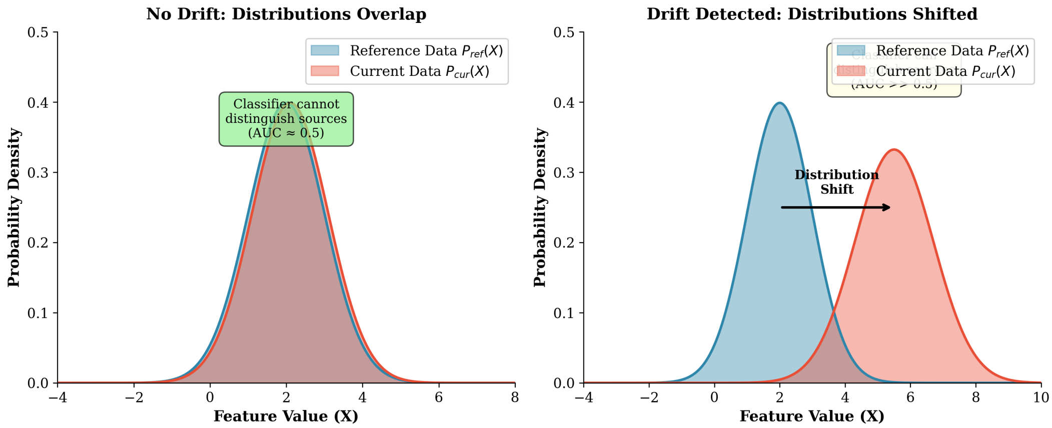

Figure 2.1: Comparison of no drift versus drift scenarios. Left: When distributions overlap, the discriminative classifier cannot distinguish data sources (AUC $\approx$ 0.5). Right: When drift occurs, distributions shift and the classifier can distinguish sources (AUC $>>$ 0.5).

2.1 Overview

Drift detection addresses the challenge that arises when the statistical characteristics of data evolve over time in a manner that impacts the validity of a machine learning model (Lu et al., 2018; Gama et al., 2014). According to Bayram et al. (2022), when users, environments, data sources, or the relationships between labels and features change, model accuracy can diminish. Gama et al. (2014) and Sculley et al. (2015) emphasize that drift detection identifies such modifications early, allowing practitioners to adapt, retrain, or update the model accordingly. A robust understanding of drift types and detection methods is essential for designing effective monitoring systems.

2.2 Historical Evolution of Drift Detection

The study of distribution drift has evolved from foundational statistical concepts to integrated AI governance over roughly seven decades. Table 2.1 summarises the major eras.

Table 2.1: Historical evolution of drift detection research.

Era	Key Developments
Early Foundations (1950s–1980s)	Statistical process control and sequential analysis laid the groundwork. Page (1954) introduced the cumulative-sum (CUSUM) change-point test, which remains foundational for sequential change detection. Reservoir sampling (Vitter, 1985) enabled representative subsampling of data streams.
Identification Era (1998–2004)	The term "concept drift" gained formal academic recognition (Widmer and Kubat, 1996). Gama et al. (2004) introduced the Drift Detection Method (DDM), using PAC-learning error-rate bounds to trigger alerts—establishing the supervised error-monitoring paradigm.
Windowing & Statistical Tests (2007–2015)	Bifet and Gavaldà (2007) proposed ADWIN (Adaptive Windowing), enabling drift detection without fixed window sizes by applying statistical tests over dynamically adjusted sub-windows. Non-parametric tests (KS, MMD) became standard for comparing reference and current distributions (Rabanser et al., 2019; Gretton et al., 2012).
Deep Learning & High-Dimensionality (2018–2023)	As models moved to unstructured data, detection shifted toward monitoring internal latent representations (Rabanser et al., 2019; Liu et al., 2020). Gözüaçık et al. (2019) introduced D3, demonstrating that discriminative classifiers could serve as powerful multivariate drift detectors. Lopez-Paz and Oquab (2017) formalised the classifier two-sample test (C2ST), providing theoretical backing for this approach.
Governance & Explainability Era (2024–2026)	Drift detection became embedded in AI governance frameworks. The focus shifted toward interpretable, statistically rigorous detection: TRIPODD (Panda et al., 2024) introduced feature-interaction hypothesis testing; E-C2ST (Pandeva et al., 2024) enabled anytime-valid sequential testing via e-values; HDPD (Feng et al., 2024) decomposed performance gaps into $P(X)$ and $P(y\|X)$ components with attributions. Cloud platforms standardised SHAP-based feature attribution monitoring (Taly et al., 2021; Amazon Web Services, 2025). Hinder et al. (2024a) exposed vulnerabilities in discriminative detectors to adversarial evasion.

This progression—from reactive error monitoring to proactive, explainable distribution analysis—provides the historical context for the present work, which integrates discriminative detection with SHAP-based attribution in the streaming setting.

2.3 Types of Drift

Drift can be categorized along two main dimensions: by the nature of the change (what component of the data-generating process is affected) and by the temporal pattern of the change (how the drift manifests over time) (Lu et al., 2018; Gama et al., 2014).

2.3.1 Drift by Nature of Change

Based on which component of the joint distribution $P(X, y)$ changes, the literature (Lu et al., 2018; Gama et al., 2014; Tsymbal, 2004) classifies drift into the following types:

Concept Drift (Real Drift): The conditional distribution $P(y|X)$ changes over time while the feature distribution $P(X)$ may or may not change (Lu et al., 2018; Žliobaitė, 2010). This represents a change in the underlying relationship between inputs and outputs—the "concept" the model has learned becomes invalid. For example, the factors that determine creditworthiness may evolve due to economic shifts, even if the applicant characteristics (features) remain similar (Gama et al., 2014). Concept drift directly degrades model performance because the learned decision boundaries no longer reflect the true input-output relationship (Lu et al., 2018).

Covariate Drift (Feature Drift / Virtual Drift): The marginal distribution of input features $P(X)$ changes over time while the conditional relationship $P(y|X)$ remains stable (Tsymbal, 2004; Shimodaira, 2000). The ground-truth concept does not change, but the model may encounter inputs outside its training distribution. For example, a fraud detection model trained on transaction patterns from one demographic may receive data from a different user population, even though the definition of fraud remains unchanged (Tsymbal, 2004). Covariate drift can degrade model performance because the model is exposed to feature regions where it may not generalize well (Shimodaira, 2000).

Prior Probability Drift (Label Shift): The marginal distribution of labels $P(y)$ changes over time while $P(X|y)$ remains stable (Tsymbal, 2004). This occurs when the prevalence of different classes changes—for instance, a seasonal increase in a particular disease affecting the class balance in medical diagnosis. This is sometimes called class imbalance shift (Lu et al., 2018).

Note on Terminology: The literature uses varied terminology (Lu et al., 2018). The term "concept drift" is sometimes used as an umbrella term for any distribution change affecting model performance (Gama et al., 2014), while some reserve it specifically for changes in $P(y|X)$ (Tsymbal, 2004). This thesis focuses primarily on covariate (feature) drift detection, as it enables early warning without requiring labels.

2.3.2 Drift by Temporal Pattern

Based on how drift manifests over time, the literature (Lu et al., 2018; Gama et al., 2014; Žliobaitė, 2010) categorizes drift as follows:

- **Abrupt (Sudden) Drift:** A rapid, discrete change where the data distribution shifts quickly from one state to another within a short time period (Gama et al., 2014).
- **Gradual Drift:** A new distribution progressively replacing the old one over an extended duration, with samples from both distributions interleaved during the transition period (Lu et al., 2018).
- **Incremental Drift:** Small but continuous changes accumulating over time, eventually resulting in a significant distributional shift (Žliobaitė, 2010).
- **Recurring (Seasonal) Drift:** Previously observed distributions reappearing periodically due to cyclical patterns, such as seasonal effects in retail or recurring user behavior patterns (Lu et al., 2018; Gama et al., 2014).

Understanding drift patterns is important for detector design: abrupt drift requires rapid response, gradual drift requires sustained sensitivity, and recurring drift benefits from mechanisms that can recognize previously seen distributions (Gama et al., 2014).

2.3.3 Summary Taxonomy

Table 2.2 consolidates the drift classification used throughout this thesis.

Table 2.2: Taxonomy of drift types referenced in this thesis.

Category	Type	Description
By Cause (Lu et al., 2018; Gama et al., 2014)	Data / Feature Drift	Change in the input distribution $P(X)$ without changing the relationship to the target.
	Concept Drift	Change in the conditional $P(y \mid X)$, invalidating learned decision boundaries.
	Prior Probability Drift	Change in the label distribution $P(y)$ while $P(X \mid y)$ is stable.
By Pattern (Gama et al., 2014; Žliobaitė, 2010)	Sudden (Abrupt)	Rapid, discrete shift between two distinct distributions.
	Gradual	Old concept progressively replaced by a new one over an extended period.
	Incremental	Small, continuous changes accumulating into a significant shift.
	Recurring (Seasonal)	Previously observed distributions reappearing cyclically.

This thesis focuses primarily on **feature (covariate) drift**—changes in $P(X)$—because it can be detected without ground-truth labels and provides early warning before model performance degrades.

2.4 Classification of Drift Detection Methods

Based on comprehensive surveys of drift detection literature (Lu et al., 2018; Gama et al., 2014; Tsymbal, 2004; Hinder et al., 2024b), drift detection methods can be classified by multiple dimensions. According to Lu et al. (2018), the most fundamental distinction is based on label availability: whether the method requires access to ground-truth labels (supervised) or operates without them (unsupervised).

2.4.1 Supervised Drift Detection Methods

According to Gama et al. (2014) and Gama et al. (2004), supervised drift detection methods require access to ground-truth labels and primarily monitor model performance or prediction errors to detect distributional changes. Gama et al. (2004)

and Baena-García et al. (2006) explain that these methods operate under the assumption that unexpected changes in error characteristics—such as increases in error rate, loss magnitude, variance, or cumulative deviation—are indicative of underlying drift.

Key supervised drift detection algorithms include:

Drift Detection Method (DDM): According to Gama et al. (2004), DDM monitors the online error rate and its standard deviation. When these statistics exceed predefined thresholds derived from binomial distribution bounds, DDM signals a warning or drift state. Gama et al. (2004) note that DDM is effective for abrupt drift but may be slow to detect gradual changes.

Early Drift Detection Method (EDDM): EDDM extends DDM by tracking the distance between consecutive classification errors rather than just the error rate (Baena-García et al., 2006). This modification improves sensitivity to gradual drift by detecting when errors cluster more closely together.

Page–Hinkley Test: This sequential analysis technique uses cumulative sums to detect changes in the mean of a monitored signal (Page, 1954). When cumulative deviation exceeds a threshold, drift is indicated.

Hoeffding Drift Detection Methods (HDDM-A, HDDM-W): These methods use Hoeffding's inequality to provide statistically grounded drift decisions with theoretical guarantees (Frias-Blanco et al., 2015). HDDM-A detects abrupt changes using moving averages, while HDDM-W is designed for gradual drift using window-based statistics.

Limitations: Supervised methods fundamentally require labeled data, which may be delayed, expensive, or unavailable in many production scenarios (Lu et al., 2018; Sethi and Kantardzic, 2017). They also react to drift after it has already caused performance degradation, rather than detecting distributional change proactively.

2.4.2 Unsupervised Drift Detection Methods

Unsupervised drift detection methods do not require ground-truth labels and instead monitor changes in data distribution, model outputs, or internal representations (Tsymbal, 2004; Hinder et al., 2024b). These methods enable earlier detection of distributional shifts—before performance degradation becomes observable—making them particularly valuable in production environments with label delays (Sethi and Kantardzic, 2017).

2.4.2.1 Statistical Distribution-Based Methods

Statistical test–based methods detect drift by directly examining changes in data distributions through hypothesis testing or distance-based measures (Rabanser et al., 2019; Raab et al., 2020). A common strategy compares a reference data window against a current data window to test whether they originate from the same distribution (Bifet and Gavaldà, 2007).

Key statistical drift detection methods include:

Adaptive Windowing (ADWIN): ADWIN dynamically maintains a sliding window and automatically shrinks it when statistically significant distributional change is detected (Bifet and Gavaldà, 2007). It provides theoretical guarantees on false positive and false negative rates and adapts its window size based on observed change.

KSWIN: This method applies the Kolmogorov–Smirnov two-sample test over sliding windows to detect changes in one-dimensional feature distributions (Raab et al., 2020). KSWIN is non-parametric and makes minimal assumptions about the data distribution.

Chi-Squared Test-Based Detectors: These use the chi-squared test to compare categorical feature distributions or discretized continuous features between time periods (Lu et al., 2018).

Kullback–Leibler (KL) Divergence and Other Divergence Measures: Methods that quantify the "distance" between probability distributions, including KL divergence, Jensen–Shannon divergence, and Hellinger distance, can all be used to measure distributional shift (Dasu et al., 2006).

Maximum Mean Discrepancy (MMD): MMD is a kernel-based two-sample test that compares distributions by mapping them into a reproducing kernel Hilbert space, capable of capturing complex multivariate distributional differences (Gretton et al., 2012).

Strengths and Limitations: Statistical tests are interpretable, theoretically grounded, and label-free (Rabanser et al., 2019). However, they can be sensitive to noise (causing false alarms), may struggle in high-dimensional spaces due to the curse of dimensionality, and typically test marginal distributions rather than joint distributions

(Lu et al., 2018; Webb et al., 2016).

Comparison of Statistical Methods with Discriminative Classification:

Table 2.3 compares popular univariate statistical methods (PSI, KS) with the multivariate discriminative classification approach used in this thesis.

Table 2.3: Comparison of Drift Detection Approaches

Method	Approach	Strengths	Limitations
PSI (Population Stability Index)	Compares binned distributions of single features; PSI > 0.25 indicates significant drift	Simple, interpretable, industry standard	Univariate only; requires binning; misses feature correlations
KS Test (Kolmogorov-Smirnov)	Non-parametric test comparing cumulative distributions per feature	No distributional assumptions; exact p-values	Univariate only; multiple testing problem with many features
Discriminative Classification	Trains classifier to distinguish old vs. new data; uses AUC as drift signal	Multivariate; captures feature interactions; single unified test	Requires classifier training; no built-in explainability (addressed by EADD)

The key distinction is that PSI and KS are **univariate** methods—they test each feature independently. When monitoring d features, this requires d separate tests, leading to the "false alarm storm" problem where the probability of at least one false positive increases with d. Discriminative classification is inherently **multivariate**, testing the joint distribution $P(X_1, X_2, \ldots, X_d)$ in a single unified check.

2.4.2.2 Model-Based Discriminative Methods

According to Pan et al. (2020), Lopez-Paz and Oquab (2017), and Sethi and Kantardzic (2017), discriminative drift detection methods train an auxiliary model to distinguish between reference and current data distributions. Lopez-Paz and Oquab (2017) explain that the core idea is to frame drift detection as a classification problem: if a classifier can accurately predict whether each instance originates from the reference or current distribution, the distributions must be distinguishably different.

Key approaches include:

Domain Classifier / Discriminative Distribution Testing: According to Pan et al. (2020) and Qian et al. (2021), this approach trains a binary classifier (e.g., logistic regression, random forest, gradient boosting) to discriminate samples from reference vs. current windows. Pan et al. (2020) note that high discriminative performance (measured by accuracy, AUC, or similar metrics) indicates significant distributional shift. The technique is sometimes called "adversarial validation" in industry practice (Pan et al., 2020).

Discriminative Drift Detector (D3): Gözüaçık et al. (2019) developed this unsupervised method that trains a classifier (logistic regression) to distinguish between old and new data windows, using AUC as the drift signal. D3 has demonstrated strong performance in recent benchmarks (Lukats et al., 2024) and represents the state-of-the-art in discriminative drift detection.

Margin Density Drift Detection (MD3): Sethi and Kantardzic (2017) proposed an alternative approach that monitors margin density—the proportion of samples near the decision boundary—to detect drift without requiring true labels.

Margin-Based and Density-Based Discriminators: According to Ditzler and Polikar (2011), these methods monitor changes in classifier margin densities or decision boundary proximity to detect distributional shifts.

Strengths: Discriminative methods can capture high-dimensional, non-linear distributional shifts that marginal univariate tests may miss (Lopez-Paz and Oquab, 2017). They require no labels for the drift detection task itself (only pseudo-labels indicating data source) and naturally aggregate information across all features (Pan et al., 2020).

Limitations: These methods require training a classifier at each monitoring step, which has computational cost; performance depends on classifier choice and hyperparameters; and interpretation of drift signals may be less intuitive than statistical test p-values (Sethi and Kantardzic, 2017).

2.4.2.3 Representation-Based Methods

Representation-based methods detect drift by monitoring changes in learned representations or embeddings rather than raw feature distributions (Rabanser et al., 2019; Liu et al., 2020):

Embedding Distribution Shift: Computing statistical distances (cosine similarity, Wasserstein distance, MMD) between reference and current embedding distributions derived from neural network intermediate layers (Rabanser et al., 2019).

Deep Learning Representation Monitoring: This approach tracks stability of learned representations across time, leveraging insights from domain adaptation and transfer learning literature (Long et al., 2016).

These methods are particularly relevant for deep learning systems where raw feature distributions may not reflect semantically meaningful changes (Liu et al., 2020).

2.4.2.4 Prediction-Based / Output-Based Methods

Prediction-based methods detect drift by monitoring changes in model outputs over time without requiring evaluation of prediction correctness (Sethi and Kantardzic, 2017; Gözüaçık et al., 2019):

Prediction Distribution Shift: Monitoring changes in the distribution of model predictions (class probabilities, regression outputs) over time (Gözüaçık et al., 2019).

Uncertainty / Entropy Monitoring: Tracking prediction entropy or uncertainty scores under the assumption that rising uncertainty reflects exposure to out-of-distribution data (Guo et al., 2017).

Confidence Monitoring: Tracking changes in model confidence levels and softmax output distributions (Guo et al., 2017).

Limitations: These methods depend on model calibration. If the model is poorly calibrated—which is common for modern deep learning models—changes in confidence may not reliably indicate drift (Guo et al., 2017; Ovadia et al., 2019).

2.4.2.5 State-of-the-Art Unsupervised Detectors (2019–2025)

Recent benchmarking by Lukats et al. (2024) provides a systematic evaluation of fully unsupervised drift detectors on real-world data streams. Their study tested numerous algorithms and identified seven that function reliably on messy, real-world data. This survey represents a critical "reality check" for the field, demonstrating that detecting change without ground-truth labels is possible but fragile.

D3 (Discriminative Drift Detector): According to Gözüaçık et al. (2019), D3 trains a logistic regression classifier to distinguish between old and new data windows, using AUC as the drift signal. Lukats et al. (2024) found D3 to be the overall winner—the most robust and accurate across different data types. However, D3 remains a "black box": it signals *that* drift occurred but provides no insight into *which* features drifted or *why*.

CSDDM (Clustered Statistical Drift Detection Method): According to Wan and Wang (2021), this method uses PCA for dimensionality reduction followed by K-Means clustering, then applies statistical tests to detect cluster shifts. Lukats et al. (2024) report it performed exceptionally well on the "lift-per-drift" metric, catching drifts with minimal false alarms.

SPLL (Semi-Parametric Log-Likelihood): Kuncheva (2013) developed this approach that fits probability density models to reference data and detects drift when new observations have anomalously low likelihood under the learned distribution. It is effective for sudden drift patterns.

IBDD (Image-Based Drift Detector): According to Souza et al. (2021), this creative approach converts numerical data streams into images and applies image comparison techniques to detect distributional changes. While highly accurate, Lukats et al. (2024) note it can be biased by its frequent reset strategy.

OCDD (One-Class Drift Detector): Gözüaçık and Can (2021) proposed using one-class SVMs to define "normal" data boundaries and flag drift when incoming data falls outside these boundaries. While sound in principle, D3 generally outperformed OCDD in benchmarks (Lukats et al., 2024).

NN-DVI (Nearest Neighbor Density Variation): According to Liu et al. (2018), this method monitors changes in local data density by tracking nearest-neighbor distances. Drift is signaled when density patterns shift significantly. It performs well for gradual density changes but can be sensitive to noise.

UCDD (Unsupervised Concept Drift Detection): Shang et al. (2020) developed a clustering-based approach that assigns pseudo-labels based on cluster membership and monitors boundary shifts. It serves as a reliable baseline in unsupervised drift detection.

BNDM (Bayesian Non-parametric Drift Monitor): This method uses a Pólya tree test to detect distributional changes without parametric assumptions. However, Lukats et al. (2024) report that BNDM struggled with small sample sizes in their benchmark, limiting its practical applicability in streaming scenarios where window sizes are constrained.

UDetect (Univariate Detection): This approach monitors standard deviation changes using Shewhart control charts. While simple and computationally efficient, Lukats et al. (2024) found it had a high failure rate (45%) in their benchmark, making it unreliable for production deployment.

Industrial Drift Detection Libraries: Beyond research algorithms, several industrial libraries provide production-ready drift detection:

- **Alibi Detect**: Open-source Python library (by Seldon) implementing multivariate drift detection via Kolmogorov–Smirnov tests, MMD, and learned kernel methods.
- **NannyML**: Open-source library providing Confidence-Based Performance Estimation (CBPE) for monitoring model performance without ground-truth labels.
- **Evidently AI**: Open-source ML monitoring framework with built-in statistical drift tests for tabular data.

Gap Identified from Benchmark Analysis: While these libraries and algorithms exist, according to Lukats et al. (2024), they share a common limitation: they typically report only *whether* drift occurred, not *where* or *why*. Statistical libraries often run separate univariate tests for each feature, causing "false alarm storms" when many features are monitored simultaneously. Although Lopez-Paz and Oquab (2017) proposed the classifier two-sample test (C2ST) framework with principled significance testing, existing streaming implementations such as D3 (Gözüaçık et al., 2019) use only fixed AUC thresholds and discard the learned classifier without exploiting it for explainability. The field lacks an integrated method that combines a statistically rigorous multivariate check with granular, feature-level explanations of drift sources.

2.4.3 Hybrid and Ensemble Methods

Hybrid approaches integrate multiple sources of information—both supervised and unsupervised signals—to improve drift detection robustness (Gama et al., 2014; Krawczyk et al., 2017):

Context-Aware Drift Detection: Combining model predictions, user context, operational signals, and metadata to compute context-weighted drift scores (Krawczyk et al., 2017).

Multi-Signal Ensemble Methods: Integrating predictions, embeddings, operational KPIs, and metadata into a unified drift score (Krawczyk et al., 2017).

Drift Detection Ensembles: Combining multiple base detectors using voting, weighting, or meta-learning to reduce both false alarms and missed detections (Krawczyk et al., 2017).

2.5 Limitations of Existing Methods

While the drift detection literature offers diverse approaches, each category has characteristic limitations:

Supervised Methods (DDM, EDDM, HDDM): These require timely labeled data, which is often delayed or unavailable in production (Lu et al., 2018). They detect drift reactively—only after performance degradation has already occurred (Sethi and Kantardzic, 2017).

Statistical Distribution Methods (ADWIN, KSWIN, KL-Divergence): These can be sensitive to noise, leading to false alarms (Rabanser et al., 2019). They may struggle in high-dimensional or correlated feature spaces and typically test marginal distributions, potentially missing joint distribution changes (Lu et al., 2018; Webb et al., 2016).

Prediction/Uncertainty-Based Methods: These rely on model calibration. Modern ML models are often poorly calibrated, especially under distribution shift, causing these signals to be unreliable (Guo et al., 2017; Ovadia et al., 2019).

Context-Aware Methods: These require pre-specification of relevant context variables, which may be incomplete, noisy, or themselves drift over time (Krawczyk et al., 2017). Misspecified context can cause missed detections or spurious alarms.

Fundamental Gap: Critically, most existing methods do not directly test whether new data is statistically distinguishable from historical data in the original feature space (Lopez-Paz and Oquab, 2017). Discriminative approaches offer a natural, model-agnostic signal of drift that does not depend on predefined contexts, labels, or calibrated uncertainties—yet this perspective has received limited systematic attention in the drift detection literature (Pan et al., 2020).

This thesis aims to bridge this gap by establishing discriminative classification as a viable, general-purpose drift detection technique and extending it with explainability through rigorous empirical evaluation and practical methodology development.

2.6 Drift Mitigation and Adaptation Strategies

Detecting drift is only the first step; production systems must also respond to it. The literature describes five principal mitigation strategies, each with distinct trade-offs (Lu et al., 2018; Gama et al., 2014; Krawczyk et al., 2017):

1. **Periodic Retraining:** The model is retrained on the latest data at fixed intervals (e.g., weekly or monthly) or triggered by drift alerts. This is the most common industrial approach because of its operational simplicity, but fixed schedules may retrain unnecessarily or react too slowly (Gama et al., 2014).
2. **Weighted Retraining:** Old and new data are combined for retraining, but recent examples receive higher weights. This preserves long-term knowledge while prioritising current patterns, and is particularly effective for gradual drift (Krawczyk et al., 2017).
3. **Ensemble Methods:** A library of models—e.g., one per season or data regime—is maintained, and predictions are combined via voting, stacking, or dynamic weighting. When drift is detected, poorly performing members are discarded and new ones trained on the current distribution (Krawczyk et al., 2017).
4. **Online / Incremental Learning:** The model is updated continuously as each new data point (or mini-batch) arrives, making it naturally adaptive to gradual and incremental drift. This is common in streaming applications such as fraud detection and recommendation systems (Gama et al., 2014; Ditzler et al., 2015).

5. **Human-in-the-Loop:** Uncertain or drifted samples are routed to human experts for re-labelling, which feeds into the next training iteration. This strategy is essential in domains where errors carry high cost (e.g., healthcare, autonomous driving) and labelling is expensive (Lu et al., 2018).

Gap in Mitigation Strategies: Existing mitigation approaches assume that the practitioner has already diagnosed the drift and selected the correct response. In practice, the choice between a targeted feature-pipeline fix, partial retraining on a subset of features, and full retraining depends on *which* features drifted and by *how much*—diagnostic information that current detectors do not provide. EADD's automated prescription system (Chapter 3) addresses this gap by mapping SHAP-based attribution patterns directly to mitigation actions.

2.7 Industry Applications of Drift Detection

Drift detection and mitigation are now deployed across multiple high-stakes domains:

Finance: Credit scoring models are particularly susceptible to "segment drift," where certain demographic groups' financial behaviours shift due to economic conditions. Qian et al. (2021) demonstrated discriminative detection for credit scoring, filtering training samples to maintain alignment between training and deployment distributions. Population Stability Index (PSI) remains the industry-standard univariate monitor for regulatory compliance.

Healthcare: Clinical prediction models degrade when patient populations change or when data acquisition technology evolves. Nestor et al. (2019) showed that ML models for common clinical tasks failed to generalise across temporally shifted cohorts within the same hospital. Vela et al. (2022) documented measurable accuracy losses within months of deployment, underscoring the need for continuous monitoring in medical AI.

Cybersecurity: Malware detection models face adversarial concept drift as attack patterns evolve to evade existing detectors. Static models become obsolete rapidly, and drift-triggered retraining is critical for maintaining detection rates against emerging threats (Ditzler et al., 2015).

E-Commerce and Recommendations: Recommendation engines must adapt to seasonal demand spikes (e.g., Black Friday, holiday periods) and evolving user preferences. Recurring drift patterns are common, and ensemble-based switching between seasonal models is a practical response (Gama et al., 2014; Krawczyk et al., 2017).

Cloud MLOps Platforms: As described in Section 2.8.2, Google Vertex AI (Taly et al., 2021) and AWS SageMaker Clarify (Amazon Web Services, 2025) have standardised feature attribution monitoring in production, signalling industry consensus that drift diagnostics must move beyond binary alarms toward explainable, feature-level attribution.

2.8 Related Work: Discriminative and Attribution Approaches

2.8.1 Discriminative Distribution Testing

Discriminative approaches to distribution comparison reframe the problem as a binary classification task: a classifier is trained to predict whether each instance originates from a reference distribution or a target distribution. High discriminative performance (AUC $\gg 0.5$) indicates the distributions are statistically separable; near-random performance suggests they are indistinguishable (Pan et al., 2020; Qian et al., 2021).

The practical need for this approach was first demonstrated at industrial scale. Pan et al. (2020) applied adversarial validation at Uber for their MaLTA user targeting system—training a classifier to detect train-deployment mismatch and triggering retraining when strong separability was found. Qian et al. (2021) applied discriminative testing to manage dataset shift in credit scoring. **D3** (Gözüaçık et al., 2019) operationalised this paradigm for streaming detection, training logistic regression classifiers on sliding windows and triggering drift when AUC exceeded a fixed threshold of 0.7. D3 was identified as the most robust unsupervised detector in the Lukats et al. (2024) benchmark.

However, D3 applies no formal significance test—its fixed AUC threshold lacks statistical justification, making it vulnerable to false alarms on autocorrelated data.

The *classifier two-sample test* (C2ST), formalised by Lopez-Paz and Oquab (2017), provided theoretical foundations for the discriminative paradigm: significance is calibrated via permutation testing, allowing principled Type I error control. The approach is label-independent, model-agnostic, and makes no assumptions about which features are important. Pandeva et al. (2024) extended C2ST with anytime-valid e-values for sequential monitoring. However, both C2ST and E-C2ST are designed for distribution testing, not streaming drift detection—and neither provides feature-level attribution of the detected shift.

Despite strong detection performance, D3 and all C2ST-based streaming detectors discard the trained classifier after computing AUC, losing the diagnostic value embedded in the classifier's learned decision boundary—what we term the "detect-and-discard" pattern. This means practitioners learn *whether* drift occurred but receive no guidance on *which* features caused it.

2.8.2 Industrial Applications of Discriminative Detection

Uber's MaLTA System: Pan et al. (2020) applied discriminative distribution testing at Uber for their MaLTA (Machine Learning Targeting Automation) user targeting system. They trained a classifier to distinguish historical training data from newly observed targeting data. When the classifier exhibited strong separability, it signaled train-deployment mismatch, prompting feature filtering and retraining adjustments before deployment.

Credit Scoring: Qian et al. (2021) applied discriminative testing to manage dataset shift in credit scoring, using the classifier's predictions to filter training samples—keeping only those most similar to the current deployment distribution.

Image Classification and Operations: Additional applications include quantifying dataset complexity in image classification (Renza and Moya-Albor, 2024) and detecting when historical data is no longer representative in manufacturing/operations settings (Senoner et al., 2023).

Feature Attribution Monitoring at Google: In an industry report, Taly et al.

(2021) described how Google's Vertex AI platform uses feature attribution monitoring to detect model degradation in production. Their system computes feature attribution baselines during model training and continuously compares production attributions against these baselines, alerting engineers with diagnostic information about which features changed. According to the report, this approach detected a service degradation incident that traditional input-distribution monitoring missed, because the drift manifested in feature importance patterns rather than in raw input statistics. While this is industry-reported evidence rather than peer-reviewed research, the deployment at Google scale suggests that feature-level attribution analysis can provide superior drift diagnostics compared to aggregate distribution tests.

Cloud Industry Standardization: The necessity of feature attribution monitoring is rapidly becoming an industry standard across major cloud providers. As with Google's Vertex AI, Amazon Web Services (AWS) has integrated "Feature Attribution Drift" monitoring into Amazon SageMaker Clarify (Amazon Web Services, 2025). SageMaker automatically calculates a SHAP baseline during training and actively monitors production data for deviations in these attribution scores. This cross-platform adoption by leading cloud providers indicates that drift detection is increasingly moving toward continuous, explainable feature attribution rather than basic distribution tracking.

2.8.3 Recent Advances in Drift Detection (2024–2026)

The drift detection landscape has evolved rapidly along three axes: statistical rigour, explainability, and computational efficiency.

Statistical Testing Advances. Pandeva et al. (2024) proposed E-C2ST, which replaces p-values with e-values in classifier two-sample tests, enabling anytime-valid sequential monitoring without multiple testing penalties. This represents a strictly superior alternative to fixed-budget permutation testing for sequential settings. Wan et al. (2024) introduced Maximum Concept Discrepancy for online drift detection using contrastive learning, offering an alternative to the discriminative paradigm.

Explainability and Attribution. Panda et al. (2024) proposed feature-interaction aware hypothesis testing with formal power guarantees, enabling identification of which specific feature interactions drive detected drift. Feng et al. (2024) introduced a hierarchical nonparametric decomposition at NeurIPS 2024 that separates covariate shift ($P(X)$) from concept shift ($P(y|X)$) with per-feature Shapley attributions and confidence intervals. Edakunni et al. (2024) decomposed performance drift into virtual and real drift components using distributional Shapley values. Decker et al. (2024) combined optimal transport with Shapley values to attribute performance degradation to specific feature shifts. Lyu et al. (2025) proposed GAM-based concept shift characterisation with knockoff-based false discovery rate control, achieving inherently interpretable shift explanations. Komnick et al. (2025) argued that only causal (not correlational) drift explanations are truly actionable, introducing causal attribution methods for drift remediation.

Formal Frameworks. Hinder et al. (2024c) formalised a rigorous mathematical definition of feature-wise drift, enabling detector-agnostic feature selection for drift localisation. A systematic literature review by Pelosi et al. (2025) confirmed that while the intersection of XAI and drift detection is rapidly growing, most existing approaches adapt static XAI methods to streaming settings without addressing the fundamental challenges of scalability and interaction-level analysis.

Label Efficiency. Abdi et al. (2024) achieved label-efficient semi-supervised detection, and Cerqueira et al. (2022) proposed student–teacher methods for unsupervised detection. Greco et al. (2024) addressed drift in unstructured data through per-label embedding distribution monitoring. Song et al. (2022) introduced autoregressive error monitoring with confidence intervals.

Robustness Concerns. Hinder et al. (2024a) demonstrated that adversarial data streams can be constructed to evade common drift detectors, including discriminative approaches, highlighting the need for robust detection mechanisms.

Key Observation: While these advances have substantially improved individual aspects of drift detection, a critical gap persists: no existing framework integrates (1) computationally efficient, statistically rigorous multivariate detection with (2) interaction-aware feature-level attribution and (3) continuous severity quantification in a unified streaming pipeline. The closest approaches—TRIPODD for interaction-aware testing, E-C2ST for sequential statistical rigour, and HDPD for attribution decomposition—each address only one dimension of this gap.

2.9 Gap in the Literature

The preceding review reveals a progressive evolution from supervised error monitoring toward unsupervised, attribution-aware drift detection. Recent advances—TRIPODD's interaction-aware testing (Panda et al., 2024), E-C2ST's sequential e-value testing (Pandeva et al., 2024), HDPD's hierarchical decomposition (Feng et al., 2024), and CDSeer's label-efficient detection (Abdi et al., 2024)—have substantially advanced individual aspects. However, four critical gaps persist:

1. **The "Black Box" Problem:** The benchmark-leading D3 (Lukats et al., 2024) and most unsupervised detectors provide binary detection signals without feature-level attribution. While TRIPODD (Panda et al., 2024) addresses interactions and HDPD (Feng et al., 2024) provides attributions, neither operates as a unified streaming detection-and-diagnosis pipeline.
2. **Marginal Attribution Blindness:** Methods that provide feature-level explanations—including standard SHAP analysis and the XPE framework (Decker et al., 2024)—typically report marginal feature contributions. They miss correlated feature drift where the interaction between features shifts while marginal distributions remain stable (Hinder et al., 2024c). Interaction-level diagnosis remains largely unexplored in the streaming drift detection context.
3. **Computational Inefficiency of Statistical Calibration:** Fixed-budget permutation testing (the standard approach for calibrating classifier two-sample tests (Lopez-Paz and Oquab, 2017)) requires a predetermined number of permutations regardless of evidence strength. When drift is obvious or clearly absent, most permutations are wasted computation. Adaptive stopping rules that maintain Type I error guarantees have been developed for Monte Carlo tests (Gandy, 2009) but have not been applied to discriminative drift detection.
4. **Absence of Continuous Severity Quantification:** Existing detectors produce binary decisions (drift/no-drift) without quantifying drift severity. Production MLOps systems require a continuous severity metric

to calibrate response urgency—distinguishing, for example, between a catastrophic sensor failure requiring immediate intervention and a gradual population shift warranting scheduled retraining.

This thesis addresses these gaps by proposing **Enhanced Adversarial Drift Detection (EADD)**, which makes three specific methodological contributions beyond D3 (Gözüaçık et al., 2019): (1) inspired by the statistical analysis of C2ST (Lopez-Paz and Oquab, 2017) and its extensions, an **Adaptive Sequential Permutation Test (ASPT)** with **prediction entropy monitoring** (H_{pred}) that provides early-warning signals while reducing computational cost and maintaining Type I error control; (2) **SHAP interaction-aware drift localisation** and a novel **Drift Severity Index (DSI)** that identifies which features caused drift and quantifies how severe the shift is; and (3) **intensive experimental evaluation** demonstrating detection coverage, false alarm control, and attribution accuracy against seven state-of-the-art detectors.

CHAPTER III
METHODOLOGY

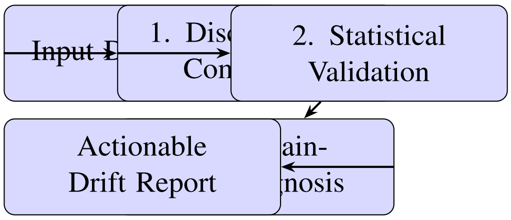

Figure 3.1: High-level overview of the EADD methodology, illustrating the four core conceptual stages.

3.1 Overview: Enhanced Adversarial Drift Detection (EADD)

This thesis proposes **Enhanced Adversarial Drift Detection (EADD)**, a framework for **feature drift** ($P(X)$ shift) detection that extends the discriminative paradigm with three methodological contributions. The fundamental insight underlying discriminative drift detection is that **feature drift detection is reducible to binary classification**: if a classifier can reliably distinguish reference data from current data based solely on features, the distributions have diverged (Lopez-Paz and Oquab, 2017; Gözüaçık et al., 2019).

While D3 (Gözüaçık et al., 2019) successfully employs this principle and Lopez-Paz and Oquab (2017) established the theoretical framework for significance calibration, the gaps identified in Chapter 2—lack of statistical rigour in D3's fixed threshold, the detect-and-discard pattern, and absence of severity quantification—remain unaddressed. EADD integrates three novel components that collectively address these limitations:

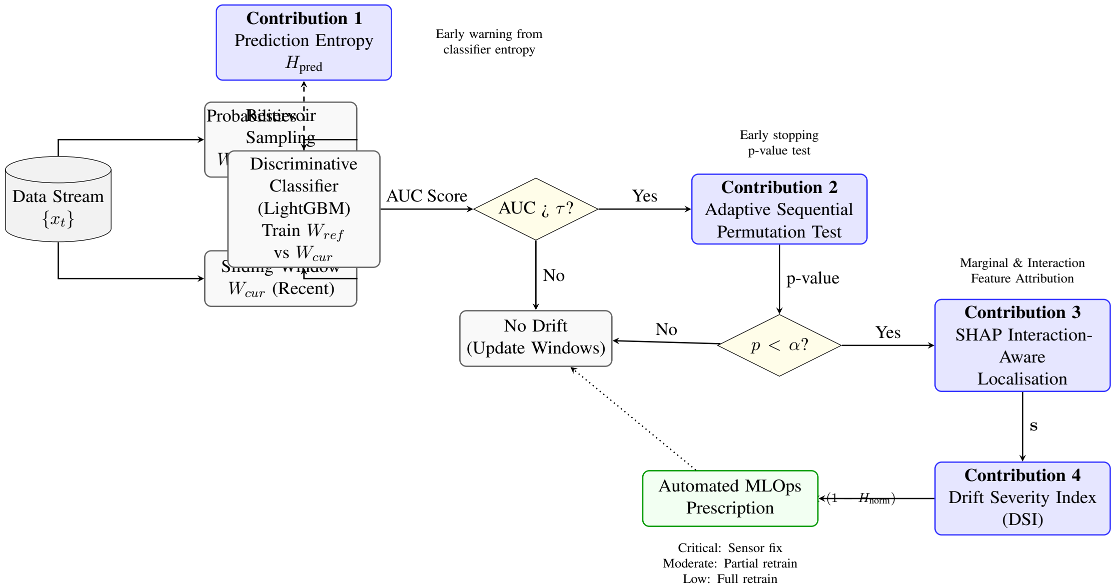

Figure 3.2: Detailed architecture of the Enhanced Adversarial Drift Detection (EADD) methodology. Blue boxes indicate novel components grouped into three thesis contributions: Steps 2b–3 (Prediction Entropy and ASPT) form Contribution 1; Steps 4a–4b (SHAP Interaction-Aware Localisation and DSI) form Contribution 2; Contribution 3 is the comprehensive experimental evaluation (not shown as a pipeline step). This pipeline transforms standard discriminative detection into an actionable MLOps monitoring system.

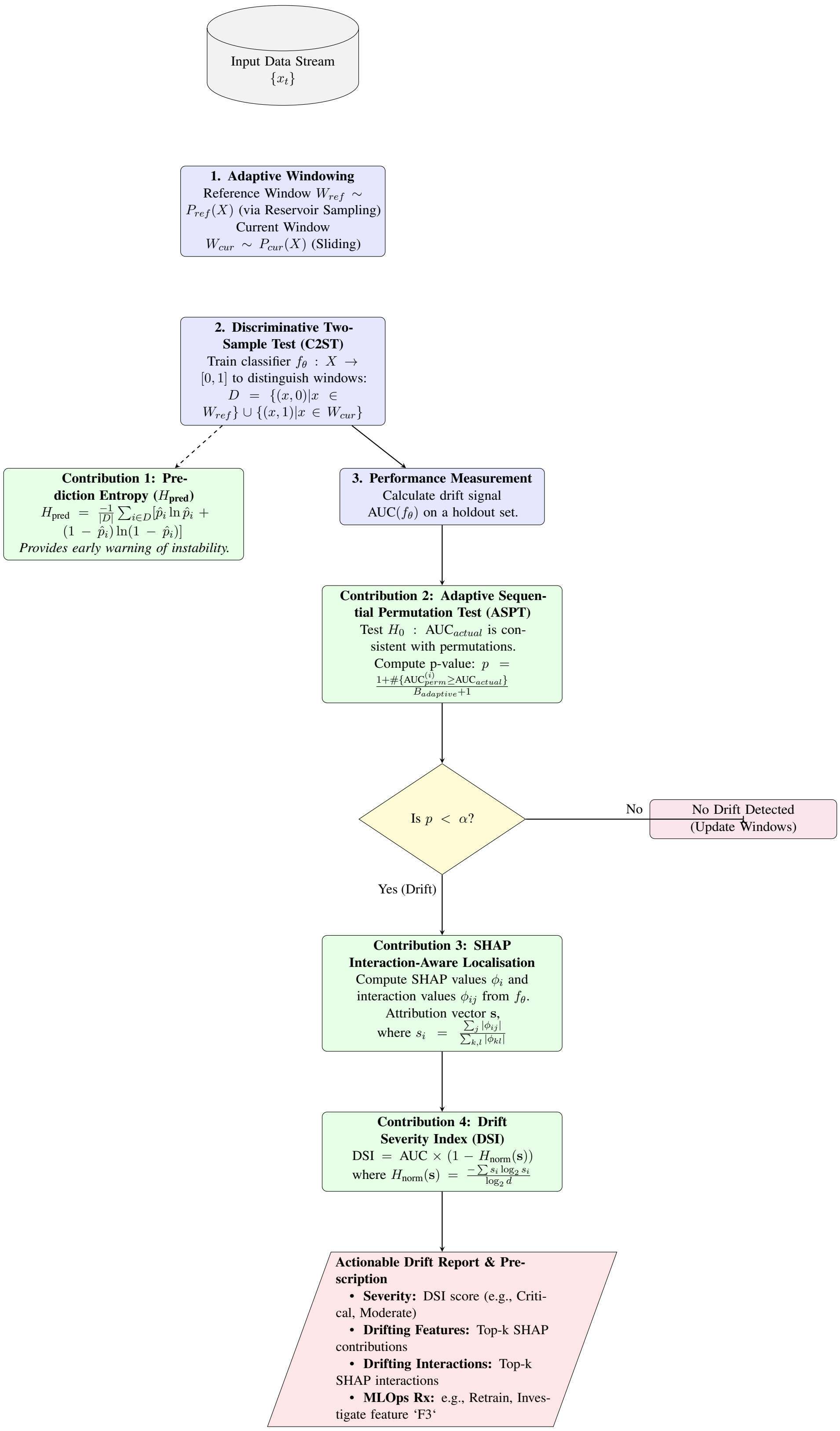

Figure 3.3: Detailed flowchart of the EADD methodology with mathematical formalisms. The process flows from data windowing (1), through discriminative classification (2) and performance measurement (3), to statistical validation via ASPT. If drift is significant, the framework proceeds to SHAP-based diagnosis and severity scoring with DSI, culminating in an actionable MLOps report and prescription. Pipeline components map to three thesis contributions: Contributions 1–2 (H_{pred} and ASPT) form Contribution 1, Contribution 3

1. Inspired by the statistical analysis of C2ST (Lopez-Paz and Oquab, 2017) and its extensions (Pandeva et al., 2024): an **Adaptive Sequential Permutation Test (ASPT)** (Gandy, 2009) with **prediction entropy monitoring** **($H_{\textbf{pred}}$)**. ASPT terminates when evidence for or against drift is sufficient, reducing computational cost by over 75% while maintaining Type I error control. H_{pred} flags subtle distributional instabilities before they reach the significance threshold.

2. Building on SHAP feature distribution analysis (Lundberg and Lee, 2017): **SHAP interaction-aware drift localisation** (Lundberg et al., 2020) and a **Drift Severity Index (DSI)**. The localisation component enables diagnosis of correlated feature drift where joint distributions shift while marginal distributions remain stable. DSI $=$ AUC $\times$ $(1 - H_{\text{norm}}(\mathbf{s}))$ combines discriminative power with attribution concentration into a continuous severity score.

3. A **comprehensive experimental evaluation** benchmarking EADD against seven state-of-the-art unsupervised detectors on synthetic and real-world streams, validating detection coverage, false alarm control, and SHAP attribution accuracy.

3.2 Problem Formulation

According to Pan et al. (2020) and Lopez-Paz and Oquab (2017), this research formalizes feature (covariate) drift detection as a problem of discriminating between historical reference data and newly arriving observations in a streaming or windowed setting. Shimodaira (2000) explains that the objective is to detect when the underlying feature distribution $P(X)$ changes substantially over time, providing an early indication of distributional drift before observable degradation in task performance occurs.

Formal Definition: Let $\{x_t\}_{t=1}^{T}$ denote a stream of feature vectors arriving over time. At each monitoring point t, a drift detector produces a binary decision $d_t \in \{\text{no-drift}, \text{drift}\}$ based on comparison between:

- A reference window W_{ref} containing historical samples presumed to represent the stable/baseline distribution
- A current window W_{cur} containing the most recent observations under test

Extended Objective: Unlike existing detectors that output only a binary drift decision, EADD additionally produces:

- A ranked list of features contributing to the drift
- Feature importance scores quantifying each feature's contribution
- An automated prescription for appropriate remediation action

According to Lu et al. (2018) and Gama et al. (2014), the detection task must balance:

- **Promptness:** Low detection delay (lag between true drift and detection)
- **Reliability:** Low false positive rate (false alarms when no drift exists)
- **Sensitivity:** High true positive rate (detecting drift when it occurs)
- **Efficiency:** Bounded computational and memory requirements suitable for production
- **Diagnostic Attribution:** Clear attribution of drift to specific features (novel contribution)

3.3 EADD Framework Components

The EADD methodology maps directly to the research requirements:

The methodology consists of five integrated steps:

Step 1: Adaptive Reference Windowing — Reservoir sampling for representative baseline

Step 2: Discriminative Classification — Train LightGBM classifier on W_{ref} vs W_{cur}

Table 3.1: Alignment of EADD Components with Research Requirements and Literature Gaps

Requirement	EADD Solution	Novelty
Discriminative Detection	Step 2: LightGBM classifier on W_{ref} vs W_{cur}	Non-linear extension of D3
Early Warning	Step 2b: Prediction Entropy (H_{pred})	Entropy of discriminator output probabilities
Statistical Rigour	Step 3: Adaptive Sequential Permutation Test	ASPT with early stopping
Feature Attribution	Step 4a: SHAP interaction values	Interaction-aware localisation
Severity Quantification	Step 4b: Drift Severity Index (DSI)	Novel composite metric

Step 2b: Prediction Entropy (H_{pred}) — Entropy-based early-warning signal (Contribution 1a)

Step 3: Adaptive Sequential Permutation Test — Evidence-driven statistical validation (Contribution 1b)

Step 4: Interaction-Aware Attribution and Severity Scoring — SHAP interaction values + DSI (Contribution 2)

3.3.1 Step 1: Adaptive Reference Windowing

To detect Feature Drift, we must compare the current data stream against a stable baseline. EADD maintains two windows consistent with established methodologies (Bifet and Gavaldà, 2007; Tsymbal, 2004):

Reference Window (W_{ref}): A buffer containing N_{ref} historical samples representing the stable feature distribution $P_{ref}(X)$. Unlike fixed sliding windows that "forget" normal data too quickly, EADD uses **Reservoir Sampling** (Vitter, 1985) to ensure the reference captures the global distribution of features, not just the most recent batches.

Current Window (W_{cur}): A buffer containing the N_{cur} most recent observations representing the current feature distribution $P_{cur}(X)$. As new data arrives, older samples are discarded to maintain responsiveness to recent changes.

Update Mechanism: At each monitoring step:

- New samples enter W_{cur}
- Oldest samples exit W_{cur}
- Reservoir sampling maintains representative W_{ref}
- Upon confirmed drift, W_{ref} may be reset to current distribution

According to Lu et al. (2018), window sizes represent a bias-variance tradeoff: larger windows provide more stable estimates but slower response; smaller windows enable faster detection but higher variance.

3.3.2 Step 2: Discriminative Classification (The Detector)

This step implements the core **classifier two-sample test** (C2ST) (Lopez-Paz and Oquab, 2017)—also referred to as "discriminative classification" or "adversarial validation" in industry practice (Pan et al., 2020)—that forms the foundation of both D3 (Gözüaçık et al., 2019) and EADD. The insight, formalised by Lopez-Paz and Oquab (2017), is that Feature Drift detection reduces to a binary classification problem: *can a model tell the difference between old and new data?*

Construct Binary Classification Dataset:

- Label all samples from W_{ref} as class 0 ("old" / reference)
- Label all samples from W_{cur} as class 1 ("new" / current)
- Combined dataset: $X = W_{ref} \cup W_{cur}$, $y = \{0\}^{N_{ref}} \cup \{1\}^{N_{cur}}$

Train Discriminative Classifier: Train a binary classifier f_θ to predict data source from features alone. EADD uses **LightGBM** (Ke et al., 2017)—a gradient boosting decision tree ensemble—as the default classifier.

Why LightGBM over D3's Logistic Regression? D3 (Gözüaçık et al., 2019) uses logistic regression, which is linear. LightGBM captures **non-linear feature interactions**, enabling detection of complex Feature Drifts where multiple features shift in correlated ways that linear models miss.

Measure Discriminative Performance: Evaluate classifier performance using stratified cross-validation:

- **AUC-ROC:** Primary metric—Area under the ROC curve

- AUC $\approx$ 0.5: Classifier cannot distinguish W_{ref} from W_{cur} $\rightarrow$ **No Feature Drift**
- AUC $>>$ 0.5: Classifier separates windows $\rightarrow P_{ref}(X) \neq P_{cur}(X) \rightarrow$ **Feature Drift detected**

Objective: If the discriminative classifier achieves high AUC, the feature distributions $P(X)$ have diverged—this is the definition of Feature Drift. Note that the term "adversarial" in EADD's name refers to this C2ST paradigm—training a classifier to distinguish two data sources—not adversarial attacks in the security or robustness sense.

3.3.3 Step 2b: Prediction Entropy Monitoring (H_{pred}) — Contribution 1a

After training the discriminative classifier (Step 2), EADD computes the **prediction entropy** H_{pred} as an early-warning signal for distributional instability. This is the normalised binary Shannon entropy (Shannon, 1948) of the C2ST discriminator's output probabilities:

$$H_{\text{pred}} = \frac{-1}{n} \sum_{i=1}^{n} [\hat{p}_i \ln \hat{p}_i + (1 - \hat{p}_i) \ln(1 - \hat{p}_i)] \tag{3.1}$$

where $\hat{p}_i = f_\theta(x_i)$ is the discriminative classifier's predicted probability that sample i belongs to the current window (W_{cur}), and $n = |W_{ref}| + |W_{cur}|$ is the combined sample count.

Interpretation:

- $H_{\text{pred}} \approx \ln 2$: Maximum entropy; the classifier cannot distinguish windows $\rightarrow$ no distributional difference
- $H_{\text{pred}} \rightarrow 0$: Low entropy; the classifier clearly separates windows $\rightarrow$ strong drift signal
- Intermediate H_{pred}: Subtle distributional change that may not yet reach statistical significance

Early Warning Property: H_{pred} captures distributional instabilities *before* they pass the ASPT significance threshold ($p < \alpha$). A declining H_{pred} trend across consecutive monitoring windows serves as a leading indicator for emerging drift, enabling proactive monitoring even when individual windows do not trigger drift alarms. This is particularly valuable for gradual and incremental drift, where the distributional change accumulates slowly.

Relationship to AUC: H_{pred} and AUC are complementary signals. AUC measures overall discriminability, while H_{pred} measures the confidence distribution of individual predictions. A high AUC with low entropy indicates clear, systematic drift; a moderately elevated AUC with higher entropy may indicate localised or emerging drift affecting only a subset of samples.

3.3.4 Step 3: Adaptive Sequential Permutation Test (ASPT) — Contribution 1b

A raw AUC score (e.g., 0.65) is inconclusive, as it may reflect genuine distributional change or statistical noise. D3 (Gözüaçık et al., 2019) relies on fixed thresholds (e.g., AUC > 0.7), which lack principled justification. Lopez-Paz and Oquab (2017) proposed the classifier two-sample test (C2ST) with significance testing, typically implemented as a fixed-budget permutation test (Good, 2005) with predetermined B permutations.

The Computational Problem: Fixed-budget permutation testing (e.g., $B = 50$) wastes computation when the evidence is clear. If the actual AUC is 0.95, the first few permutations already establish significance; the remaining permutations add cost without information. Conversely, if the AUC is 0.52, early permutations quickly reveal non-significance.

EADD's Solution — Adaptive Sequential Permutation Testing: EADD introduces an **adaptive sequential permutation test** inspired by Gandy (2009)'s sequential Monte Carlo framework. Instead of committing to a fixed budget B, ASPT monitors evidence accumulation after each permutation and terminates early when the decision boundary is crossed:

1. **Initialise:** Set maximum budget $B_{\max}$, nominal significance level α, and minimum sequential budget $B_{\min}$.
2. **Sequential Execution:** For each permutation $i = 1, \ldots, B_{\max}$:

- Shuffle labels, retrain the discriminative classifier, compute $\text{AUC}_{\text{perm}}^{(i)}$
- Update running count: $c_i = \#\{j \leq i : \text{AUC}_{\text{perm}}^{(j)} \geq \text{AUC}_{\text{actual}}\}$
- **Early reject H_0 (confirm drift):** If $c_i = 0$ and $i \geq B_{\min}$ and the conservative bound $1/(i+1) < \alpha$ holds, terminate and confirm drift with $p = 1/(i+1)$.
- **Early accept H_0 (no drift):** If $c_i/i > 2\alpha$, the running exceedance rate far exceeds the nominal level; terminate and declare no drift.

3. **Terminal decision:** If $B_{\max}$ is reached, compute the standard empirical p-value: $p = (c_{B_{\max}} + 1)/(B_{\max} + 1)$. Drift is confirmed if $p < \alpha$.

Statistical Validity Note: ASPT preserves the standard permutation p-value at termination, either by the full-budget estimate $p = (c_B + 1)/(B + 1)$ or by the conservative early-reject upper bound $p = 1/(i+1)$ when no exceedances are observed. This keeps the stopping rule aligned with the permutation-test interpretation.

Computational Savings: ASPT logs the number of permutations executed per invocation. The empirically measured mean is substantially below the maximum budget in clear no-drift cases (early acceptance), while ambiguous cases near the decision boundary consume a larger fraction. The full budget is reached only when additional permutations are most informative—keeping average cost well below the fixed-budget baseline.

3.3.5 Step 4a: Interaction-Aware Feature Drift Localisation via SHAP — Contribution 2a

Once the ASPT confirms drift ($p < \alpha$), the next objective is to determine *which features and feature interactions drifted.* D3 (Gözüaçık et al., 2019) discards the trained classifier after computing AUC—what we term the “detect-and-discard” pattern. However, the discriminative classifier has learned precisely which features—and which feature combinations—differentiate the reference data from the current data.

Per-Feature SHAP Attribution: EADD first applies **TreeSHAP** (SHapley Additive exPlanations for tree ensembles) (Lundberg and Lee, 2017; Lundberg et al., 2020) to the discriminative classifier. TreeSHAP computes exact Shapley values (Shapley,

1953) in polynomial time for tree-based models. *Per-feature* importances are computed by averaging each feature's individual SHAP contribution across all samples—capturing how much each feature, on its own, drives the classifier's ability to distinguish reference from current data:

1. Compute SHAP values using TreeSHAP (efficient for LightGBM)
2. Calculate mean absolute SHAP value per feature: $\text{Importance}_i = \frac{1}{n}\sum_{j=1}^{n}|\text{SHAP}_{ij}|$
3. Normalise to a contribution vector: $s_i = \frac{\text{Importance}_i}{\sum_k \text{Importance}_k}$, so $\mathbf{s} = (s_1, \ldots, s_d)$ sums to 1
4. Rank features by s_i; the contribution vector $\mathbf{s}$ also feeds directly into the DSI in Step 4b

SHAP Interaction Values — Addressing the Per-Feature Attribution Blind Spot: Per-feature SHAP values capture each feature's individual contribution but can miss *correlated feature drift*—scenarios where the joint distribution of feature subsets shifts while marginal distributions remain relatively stable (Hinder et al., 2024c). EADD extends attribution to **SHAP interaction values** (Lundberg et al., 2020):

1. Compute the SHAP interaction matrix $\Phi \in \mathbb{R}^{d\times d}$ (where d is the number of input features) using TreeSHAP's interaction mode (Lundberg et al., 2020), where Φ_{ij} captures the joint effect of features i and j on the discriminative classifier's prediction
2. Compute interaction importance: $\text{Interaction}_{ij} = \frac{1}{n}\sum_{k=1}^{n}|\Phi_{ij}^{(k)}|$ for all pairs $i \neq j$
3. Identify significant interactions: pairs where Interaction_{ij} exceeds a threshold defined as $\mu + 2\sigma$ of the interaction importance distribution
4. Report both marginal feature drift contributions and significant feature interaction drift

Why Interaction Values Matter: Consider a production scenario where features *Temperature* and *Humidity* individually show modest marginal drift (each contributing $\sim$15%), but their *interaction* has shifted dramatically (e.g., the correlation structure changed from $\rho = 0.3$ to $\rho = 0.8$). Marginal SHAP would report distributed, low-severity drift; SHAP interaction values correctly identify the correlated shift as the primary driver, enabling targeted diagnosis of the upstream data source that jointly affects both features.

3.3.6 Step 4b: Drift Severity Index (DSI) — Contribution 2b

Existing detectors produce binary decisions (drift/no-drift). Production systems require a continuous severity metric to calibrate response urgency. EADD introduces the **Drift Severity Index (DSI)**, a composite metric that simultaneously captures detection confidence and drift concentration:

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s})) \tag{3.2}$$

where $\mathbf{s} = (s_1, s_2, \ldots, s_d)$ is the normalised SHAP contribution vector, d is the number of input features, and $H_{\text{norm}}(\mathbf{s})$ is the normalised Shannon entropy (Shannon, 1948):

$$H_{\text{norm}}(\mathbf{s}) = \frac{-\sum_{i=1}^{d} s_i \log_2 s_i}{\log_2 d} \tag{3.3}$$

Interpretation:

- **AUC component** captures the overall discriminability between reference and current distributions (detection confidence).
- $(1 - H_{\text{norm}})$ **component** captures the *concentration* of drift attribution. When drift is concentrated in one feature (univariate), entropy is low and $(1 - H_{\text{norm}})$ is high, yielding a high DSI. When drift is distributed uniformly across all features (multivariate shift), entropy is maximal and $(1 - H_{\text{norm}}) \approx 0$, yielding a low DSI despite high AUC.

DSI Properties:

- DSI $\in [0, 1]$: bounded, interpretable range
- DSI $\to 1$ when AUC is high and drift is concentrated (severe univariate drift / sensor failure)

- DSI $\to 0$ when AUC is low (no drift) or when drift is uniformly distributed (diffuse multivariate shift)
- DSI is monotonically increasing in AUC and monotonically decreasing in attribution entropy

DSI-Based Severity Tiers:

Table 3.2: Drift Severity Index (DSI) Tiers and Automated Prescriptions

DSI Range	Severity	Drift Pattern	Prescription
> 0.6	Critical	Concentrated drift (sensor failure, pipeline break)	Immediate data quality check; disable affected feature
0.3–0.6	Moderate	Subset drift (correlated features)	Investigate shared data source; partial retraining
< 0.3	Low	Diffuse multivariate shift	Scheduled full retraining; expand training data

This replaces the heuristic thresholds (50% for univariate, 70% for subset) used in prior work with a continuous, theoretically motivated metric that naturally adapts to the dimensionality and correlation structure of the data.

Justification for Attribution-Based Diagnosis: The choice of SHAP-based feature attribution over traditional distribution-based monitoring is supported by convergent industry adoption. Google Vertex AI uses feature attribution monitoring that detected degradation missed by input-distribution monitoring alone (Taly et al., 2021), and AWS SageMaker Clarify integrates SHAP-based attribution drift monitoring as standard capability (Amazon Web Services, 2025). Recent academic work reinforces this direction: Decker et al. (2024) demonstrated that optimal-transport-aligned Shapley attribution captures model-relevant drift more effectively than aggregate distribution tests, and Komnick et al. (2025) argued that attribution-based explanations are prerequisites for truly actionable drift remediation.

3.3.7 Automated Drift Prescription via DSI

Beyond diagnosis, EADD provides **severity-proportional remediation recommendations** driven by the Drift Severity Index (Table 3.2). Unlike prior approaches that rely on fixed heuristic thresholds, DSI-based prescriptions are calibrated by both detection confidence (AUC) and drift concentration (normalised SHAP entropy), enabling proportional responses:

Critical Severity (DSI > 0.6): Concentrated drift in one or two features, typically indicating sensor failure or data pipeline disruption.

- Immediate data quality investigation for the dominant feature(s)
- Disable affected feature(s) pending resolution
- If SHAP interaction values identify a correlated pair: investigate the shared upstream data source

Moderate Severity (DSI 0.3–0.6): Subset drift across a correlated feature group, typically indicating a systematic change in one data domain.

- Investigate common data source for the affected feature group
- Partial retraining with emphasis on the affected feature subspace
- Review feature engineering for the affected domain

Low Severity (DSI < 0.3): Diffuse multivariate shift distributed across many features, typically indicating a population-level distributional change.

- Schedule full model retraining
- Expand training data to incorporate the recent distribution
- Review deployment assumptions and feature engineering

3.3.8 Model Adaptation Strategies

According to Gama et al. (2014), upon drift detection, the primary prediction model can be adapted:

Strategy 1: Drift-Triggered Retraining

- Model is trained at initialization
- Retraining occurs only when drift alarm is raised
- Reference window resets to current data after retraining

- According to Pan et al. (2020), advantage: Conserves resources; exploits early detection for timely updates

Strategy 2: Periodic Retraining (Baseline)

- Model is retrained at fixed intervals regardless of drift signals
- Advantage: Simple; does not depend on detector reliability
- According to Gama et al. (2014), disadvantage: May retrain unnecessarily or too late

Strategy 3: Continuous Update with Drift Weighting

- Online learning with sample weighting based on drift score
- According to Krawczyk et al. (2017), recent samples receive higher weights when drift is detected

3.4 Datasets

Following the comprehensive benchmark by Lukats et al. (2024), experiments used both synthetic and real-world data streams to evaluate EADD under controlled and realistic conditions.

Synthetic Datasets: Custom synthetic data generators were used to create controlled $P(X)$ shifts with known ground-truth drift points, enabling precise measurement of detection delay and accuracy:

- **SineClusters:** Sine-modulated cluster data with controlled drift patterns
- **WaveformDrift2:** Waveform generator with gradual feature drift
- **Custom Generators:** Streams of $n = 10{,}000$ samples with $d = 5$ features and controlled abrupt, gradual, incremental, and recurring drift injected at known positions (e.g., $t = 5{,}000$)

Real-World Benchmark Datasets: Eleven data streams from the Lukats et al. (2024) benchmark were used, spanning diverse domains and drift characteristics:

- **Electricity (Elec2)** (Harries, 1999): 45,312 samples × 8 features; electricity pricing data with natural market fluctuations
- **INSECTS Variants** (Souza et al., 2020): Five variants—Abrupt Balanced, Gradual Balanced, Incremental Balanced, Incremental-Abrupt Balanced, Incremental-Reoccurring Balanced—each containing ∼24,000–80,000 samples × 33 features with documented ground-truth drift points
- **NOAA Weather:** Weather measurements with temporal drift patterns
- **Outdoor Objects:** Object recognition features with environmental variation
- **Powersupply:** Power consumption data with usage pattern changes
- **SineClusters:** Sine-modulated cluster data with controlled drift patterns (3 annotated drift points)
- **WaveformDrift2:** Waveform generator with gradual feature drift across 21 features (3 annotated drift points)

All datasets were processed in streaming fashion, with features used for drift detection. For datasets with known drift points (e.g., INSECTS variants with drift indices at specific sample positions), ground truth was used for computing detection metrics. The INSECTS datasets were particularly valuable as they provide exact drift timestamps, enabling precise measurement of detection delay and false alarm rates.

3.5 Baseline Drift Detectors

EADD was benchmarked against seven state-of-the-art unsupervised drift detectors evaluated in the Lukats et al. (2024) survey, all of which operate without ground-truth labels:

- **D3 (Discriminative Drift Detector)** (Gözüaçık et al., 2019): Trains a logistic regression classifier to distinguish old from new data windows, using AUC as the drift signal. Identified as the overall best-performing unsupervised detector in the Lukats et al. (2024) benchmark.

- **BNDM (Bayesian Non-parametric Drift Monitor):** Uses a Pólya tree test to detect distributional changes without parametric assumptions.
- **CSDDM (Clustered Statistical Drift Detection Method)** (Wan and Wang, 2021): Applies PCA dimensionality reduction followed by K-Means clustering, then uses statistical tests to detect cluster shifts.
- **IBDD (Image-Based Drift Detector)** (Souza et al., 2021): Converts numerical data streams into images and applies image comparison techniques to detect distributional changes.
- **OCDD (One-Class Drift Detector)** (Gözüaçık and Can, 2021): Uses one-class SVMs to define "normal" data boundaries and flags drift when incoming data falls outside these boundaries.
- **SPLL (Semi-Parametric Log-Likelihood)** (Kuncheva, 2013): Fits probability density models to reference data and detects drift when new observations have anomalously low likelihood.
- **UDetect (Univariate Detection):** Monitors standard deviation changes using Shewhart control charts.

Existing benchmark results for these seven detectors from the Lukats et al. (2024) framework were used for direct comparison with EADD's detection performance across the same datasets.

3.6 Evaluation Metrics

Following Lu et al. (2018), Lukats et al. (2024), and Souza et al. (2020), the following metrics were used to evaluate drift detection performance:

Detection Performance Metrics:

- **Mean Time to Detection (MTD):** Average number of samples between true drift occurrence and detection alarm. Lower is better.
- **Missed Detection Rate (MDR):** Proportion of true drifts that were not detected within a tolerance window. Lower is better.

- **Mean Time between False Alarms (MTFA):** Average number of samples between consecutive false alarms. Higher is better.
- **Mean Time to Recovery (MTR):** Average time for the system to stabilize after drift detection.
- **F1-Score:** Harmonic mean of precision and recall for drift detection decisions.
- **False Positive Rate (FPR):** Proportion of alarms when no drift exists. Lower is better.

Explainability Metrics:

- **Feature Attribution Accuracy:** Whether SHAP correctly identifies the ground-truth drifting feature(s) as the top contributor(s).
- **SHAP Dominance Score:** Percentage of total SHAP importance assigned to the true drifting feature(s). Higher indicates more precise attribution.
- **Prescription Accuracy:** Whether the automated prescription matches the actual drift scenario (univariate, subset, or multivariate).

3.7 Experimental Design

Four experiments were designed to systematically validate EADD's detection accuracy, SHAP attribution accuracy, and robustness:

Experiment 1: Sensitivity to Temporal Drift Patterns

Evaluated EADD's ability to detect all four major temporal drift patterns (abrupt, gradual, incremental, recurring) on synthetic data with known ground-truth drift points. Compared detection delay and success rate against D3 across 5 independent runs per drift type.

Experiment 2: Real-World Benchmark Comparison

Benchmarked EADD against seven unsupervised baselines (D3, BNDM, CSDDM, IBDD, OCDD, SPLL, UDetect) across 11 benchmark data streams from the Lukats et al. (2024) framework, measuring MTD, MDR, MTR, and overall F1.

Experiment 3: Explainability Case Study

Validated SHAP-based feature attribution accuracy across three controlled synthetic scenarios: (1) univariate drift in a single feature, (2) correlated subset drift across 2–3 features, and (3) multivariate drift distributed across all features. Measured whether EADD correctly identifies drifting features and generates appropriate prescriptions.

Experiment 4: Robustness to Stable Data (False Alarm Evaluation)

Assessed false alarm rates on four types of stable synthetic streams with zero actual drift (Gaussian i.i.d., autocorrelated, heteroscedastic, correlated features). Evaluated EADD's permutation test robustness and compared against D3 at multiple AUC thresholds $\tau \in \{0.6, 0.7, 0.8\}$.

3.8 Practical Considerations

Class Imbalance: When $|W_{ref}| \neq |W_{cur}|$, use:

- Stratified cross-validation
- Class-weighted classifiers
- According to Lopez-Paz and Oquab (2017), AUC-ROC rather than accuracy

Computational Efficiency:

- Use efficient classifiers (logistic regression, small random forests)
- Consider approximate methods for large windows
- According to Pan et al. (2020), implement incremental training where possible

Temporal Leakage Prevention:

- Use purged cross-validation to avoid temporal overlap between folds
- According to Souza et al. (2020), ensure strict temporal ordering in train/test splits

Threshold Calibration:

- Use validation data with known drift to tune T_{drift}
- According to Pan et al. (2020), consider adaptive thresholds based on baseline variance

CHAPTER IV
EXPERIMENTS AND RESULTS

This chapter presents the experimental evaluation of Enhanced Adversarial Drift Detection (EADD). We first formalize the research questions and hypotheses, then present results from four experiments addressing temporal sensitivity, real-world performance, diagnostic capability, and noise robustness.

Note on experimental protocols. Two synthetic experiments evaluate EADD on the same four drift types. Experiment 1 (Section 4.4) uses a two-detector temporal-pattern protocol (EADD vs D3, n=10,000). Experiment 5, reported in the hypothesis-testing section (Section 4.7), compares all eight detectors (n=5,000). Both experiments confirm that EADD detects all four drift types (4/4) with zero false alarms—the only detector to achieve this combination.

4.1 Research Questions

Based on the gaps identified in Chapter 2, this thesis investigates three research questions:

- **RQ1 (Detection Accuracy):** Does EADD achieve comparable or superior drift detection performance (F1-score, detection delay) to state-of-the-art unsupervised detectors across synthetic and real-world data streams?
- **RQ2 (Explainability):** Can EADD's SHAP-based feature attribution correctly identify the root cause of drift, and does the automated prescription system provide accurate drift categorization?
- **RQ3 (Robustness):** Does EADD's permutation test significantly reduce false alarm rates on stable data compared to threshold-based approaches like D3?

4.2 Hypotheses

Three formal hypotheses were tested:

- **H1 (Detection Coverage):** EADD detects a broader range of temporal drift patterns than D3. Formally: H1a tests whether EADD detects significantly more drift types (binomial test), and H1b tests whether EADD achieves comparable detection delay where both detectors succeed (Mann–Whitney U test, $\alpha = 0.05$).
- **H2 (False Alarm Reduction):** EADD produces significantly fewer false alarms than D3 on stable data. Formally, H_0: $\text{FPR}_{\text{EADD}} \geq \text{FPR}_{\text{D3}}$ vs H_1: $\text{FPR}_{\text{EADD}} < \text{FPR}_{\text{D3}}$. Tested using the Mann–Whitney U test (one-sided, $\alpha = 0.05$) at multiple D3 threshold settings ($\tau \in \{0.6, 0.7, 0.8\}$).
- **H3 (Explainability Accuracy):** EADD correctly identifies the ground-truth drifting feature(s) as the top SHAP contributor in controlled experiments. Evaluated qualitatively across three controlled synthetic scenarios ($\alpha = 0.05$).

4.3 Experimental Setup

Environment: Experiments were conducted in Python 3.10 using LightGBM (Ke et al., 2017) for the discriminative classifier and SHAP (SHapley Additive exPlanations) for feature attribution (Lundberg and Lee, 2017). Implementation is available in the `detectors/eadd.py` module within the Lukats et al. (2024) benchmark framework.

EADD Configuration:

- **Reference Window (N_{ref}):** 500 samples
- **Current Window / Step Size (N_{cur}):** 200 samples
- **AUC Threshold:** 0.7 (adapted from industry practice (Pan et al., 2020))
- **ASPT:** $B_{\max} = 50$ permutations, $B_{\min} = 10$, $\alpha = 0.05$ (adaptive early stopping)

- **LightGBM:** 50 estimators, learning rate 0.1, max depth $= -1$ (unlimited), 31 leaves
- **Cross-Validation:** Stratified 5-fold within each detection cycle
- **Monitoring Frequency:** Every 50 samples

D3 Baseline Configuration:

- Logistic regression classifier
- Matching window configuration ($N_{ref} = 500$, $N_{cur} = 200$)
- AUC threshold of 0.7 (no permutation test)

Baselines: EADD was compared against seven unsupervised detectors from the Lukats et al. (2024) benchmark: D3 (Gözüaçık et al., 2019) (primary comparison), BNDM, CSDDM, IBDD, OCDD, SPLL, and UDetect. All baselines were run with their default configurations as provided in the benchmark framework. A per-detector timeout of 120 seconds was applied for real-world datasets to handle scalability limitations of certain detectors (e.g., BNDM on high-dimensional data).

4.4 Experiment 1: Multi-Detector Sensitivity to Temporal Drift Patterns

Objective: Systematically compare EADD against seven state-of-the-art unsupervised drift detectors across all four major temporal drift patterns—abrupt, gradual, incremental, and recurring—to evaluate the trade-offs between detection speed, missed drift rate, and false alarm rate.

Data Generation: Synthetic data streams of $n = 10{,}000$ samples with $d = 5$ features were generated for each drift type. Drift was injected at known points as follows:

- **Abrupt:** Instantaneous mean shift of $+2\sigma$ at $t = 5{,}000$
- **Gradual:** Linear interpolation between old and new distributions over 2,000 samples
- **Incremental:** Small step increases of 0.1σ every 200 samples

- **Recurring:** Oscillating between original and shifted distributions with period 2,000 (3 drift points)

All eight detectors were run under identical conditions with their default configurations from the Lukats et al. (2024) benchmark framework.

Baselines: D3 (Gözüaçık et al., 2019), BNDM, CSDDM, IBDD, OCDD, SPLL, and UDetect.

Results:

Table 4.1 presents the detection performance across all eight detectors and four drift types.

Table 4.1: Experiment 1: Multi-Detector Comparison on Synthetic Drift Patterns. MTD = Mean Time to Detection (samples); MDR = Missed Detection Rate; FA = False Alarms. "—" indicates the detector missed all drift points.

	Abrupt			**Gradual**			**Incremental**			**Recurring**		
Detector	MTD	MDR	FA	MTD	MDR	FA	MTD	MDR	FA	MTD	MDR	FA
EADD	149	0.0	**0**	699	0.0	**0**	849	0.0	**0**	149	0.0	**0**
D3	**98**	0.0	0	—	1.0	—	—	1.0	—	157	0.0	0
BNDM	272	0.0	4	**26**	0.0	6	383	0.0	6	219	0.0	3
CSDDM	206	0.0	5	413	0.0	6	**9**	0.0	5	84	0.0	6
IBDD	**18**	0.0	41	153	0.0	48	156	0.0	47	**11**	0.0	49
OCDD	249	0.0	18	249	0.0	18	249	0.0	18	249	0.0	16
SPLL	66	0.0	0	—	1.0	—	—	1.0	—	72	0.0	0
UDetect	—	1.0	—	—	1.0	—	—	1.0	—	—	1.0	—

Figure 4.1 presents heatmaps of detection performance across all detector–drift-type combinations, providing a visual summary of Table 4.1.

Figure 4.2 shows the feature distribution bell curves before and after drift injection, visualizing the distributional shift that each detector must detect.

Figure 4.3 shows the multi-detector time-series comparison, plotting each detector's AUC or detection score over time alongside the true drift point.

Key Findings: The multi-detector comparison reveals three distinct detector categories:

1. **EADD Achieves Unique Coverage–Precision Balance:** EADD is the only detector that detected all four drift types (4/4 coverage, MTD: 149–

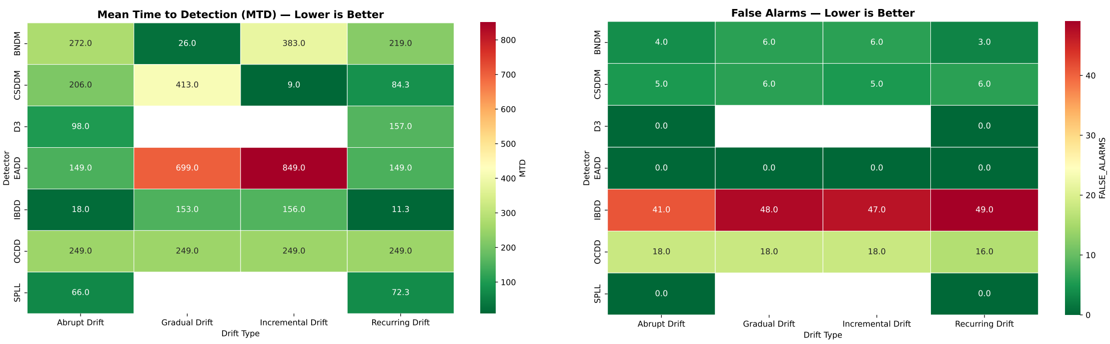

Figure 4.1: Heatmaps of detection performance on synthetic drift. Left: Mean Time to Detection (lower is better; white cells indicate complete miss). Right: False Alarm count (lower is better).

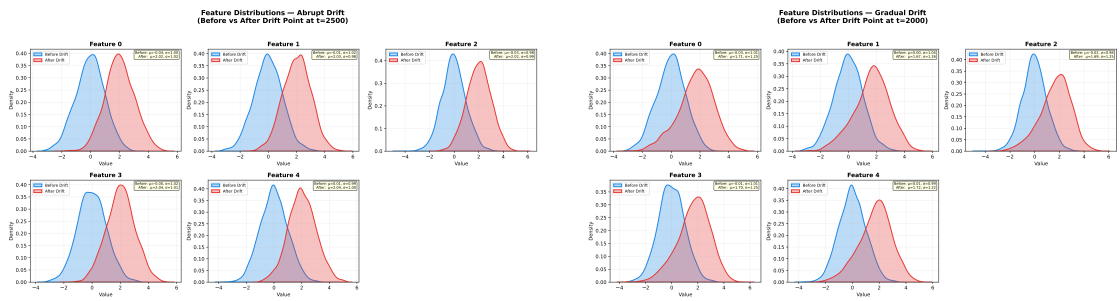

Figure 4.2: Bell curves showing feature distributions before (blue) and after (red) drift injection. Left: Abrupt drift with clear separation. Right: Gradual drift with overlapping distributions—the subtle shift that D3, SPLL, and UDetect all failed to detect.

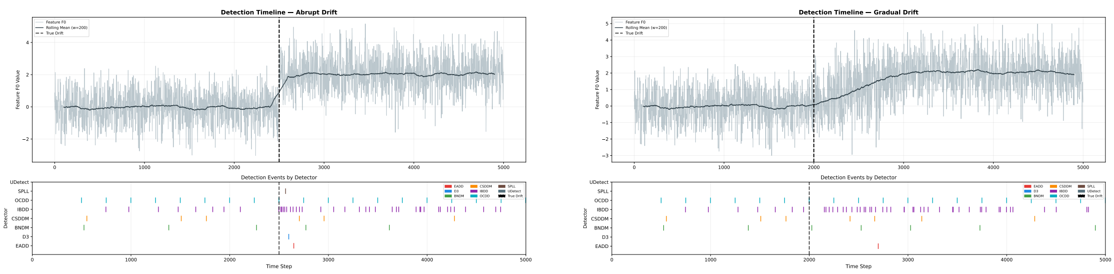

Figure 4.3: Time-series comparison of all eight detectors. Left: Abrupt drift—most detectors respond near the true drift point (dashed line). Right: Gradual drift—only EADD, BNDM, CSDDM, IBDD, and OCDD detect the shift; D3, SPLL, and UDetect remain silent.

849) with zero false alarms. Its two-stage statistical validation—AUC threshold screening followed by ASPT permutation testing ($p < 0.05$)—eliminates spurious detections while retaining sensitivity to all temporal drift patterns, including gradual and incremental drift that D3 and SPLL missed entirely. EADD's detection delays are longer than the fastest detectors (e.g., IBDD's 11–18 samples), reflecting the cost of statistical rigour; however, each EADD detection is accompanied by actionable SHAP attribution and DSI severity quantification that no other evaluated detector provides.

2. **High-Sensitivity Detectors with False Alarm Costs:** BNDM, CSDDM, IBDD, and OCDD also achieved 4/4 drift-type coverage, but with substantially higher false alarm rates (3–49 per drift type). IBDD was the fastest (MTD of 11–18 samples) but generated 41–49 false alarms per type, rendering it impractical for production monitoring. OCDD produced 16–18 false alarms per type with no speed advantage over EADD.

3. **Complete Failures:** D3 and SPLL both missed gradual and incremental drift entirely (MDR = 1.0), detecting only 2/4 types. UDetect failed on all four types.

False Alarm Robustness. To evaluate ASPT's robustness on stable data, four types of stable synthetic streams with zero actual drift were generated: (1) Gaussian i.i.d., (2) autocorrelated AR(1), (3) heteroscedastic, and (4) correlated features ($\rho = 0.7$). Each stream was processed 5 times by both EADD and D3 at multiple thresholds. EADD produced zero false alarms across all four stream types. D3 was particularly vulnerable to autocorrelated data (87.4 false alarms at τ=0.6, 54.6 at τ=0.7). The ASPT correctly identifies temporal autocorrelation as non-significant ($p > 0.05$) because permuted AUC scores match the observed AUC by chance. A Mann–Whitney U test confirmed that EADD produces significantly fewer false alarms than D3 at τ=0.6 ($p = 0.0101$). These results are consistent with those reported in Experiment 2 across real-world stable streams.

4.5 Experiment 2: Multi-Detector Real-World Benchmark

Objective: Evaluate all eight detectors on real-world data streams from the Lukats et al. (2024) benchmark where drift patterns are unknown, complex, and multi-dimensional.

Procedure: Eleven real-world datasets from the Lukats et al. (2024) benchmark were processed in streaming fashion. All eight detectors were applied with their default configurations. For datasets with known ground-truth drift points (InsectsAbrupt, InsectsGradual, SineClusters, WaveformDrift2), standard detection metrics (MTD, MDR) were computed. For datasets without ground truth, the number of detections was recorded.

A per-detector timeout of 120 seconds was applied, as BNDM's Bagged-NearestDelta computation exceeded practical limits on high-dimensional (33-feature) datasets.

Results:

Table 4.2 presents the results on datasets with known drift points.

Table 4.2: Experiment 2: Multi-Detector Benchmark on Datasets with Known Drift Points. MTD = Mean Time to Detection (samples). Bold indicates best MTD. "—" indicates detector missed all drifts or timed out.

Detector	InsectsAbrupt	InsectsGradual	SineClusters	WaveformDrift2
EADD	148	—	299	**99**
D3	153	—	219	179
BNDM	—	—	499	60
CSDDM	**44**	**143**	**35**	359
IBDD	28	88	45	**8**
OCDD	254	—	249	249
SPLL	**25**	39	249	22
UDetect	—	—	36	21

Table 4.3 summarizes detection activity on datasets without annotated ground-truth drift points.

Figure 4.4 presents heatmaps summarizing detection performance across all detector–dataset combinations on the real-world benchmark.

Figure 4.5 shows time-series visualizations of detection behavior on two representative real-world datasets.

Table 4.3: Experiment 2: Detection Counts on Datasets Without Ground Truth (lower is generally preferred, as excessive detections suggest false alarms). “TO” = timed out.

Dataset	EADD	D3	BNDM	CSDDM	IBDD	OCDD	SPLL
Electricity	39	37	40	79	726	37	75
IncrAbrupt	1	0	2*	15	94	79	0
Incremental	1	0	1*	19	63	79	1
Reoccurring	2	2	3*	21	107	79	1
NOAAWeather	13	29	36	70	485	71	2
OutdoorObjects	6	1	8	15	34	15	14
Powersupply	7	11	16	58	163	79	9
Total	**69**	**80**	**106***	**277**	**1,672**	**439**	**102**

*BNDM timed out (120s) on all 33-feature INSECTS datasets; counts reflect detections before timeout.

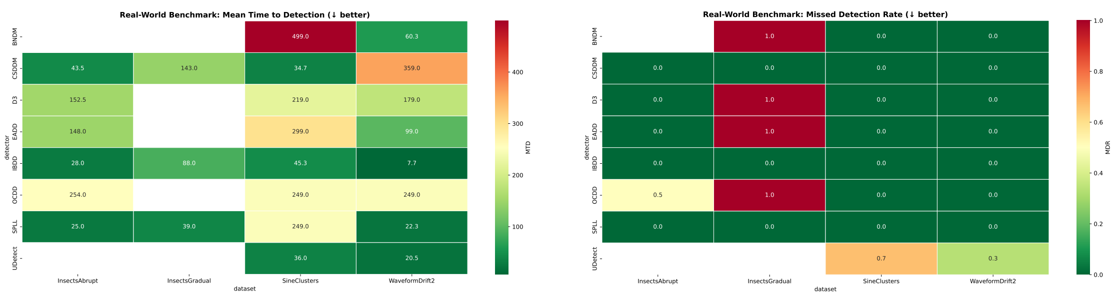

Figure 4.4: Heatmaps of real-world benchmark performance. Left: Mean Time to Detection (lower is better; white cells indicate missed drifts or timeout). Right: Missed Detection Rate (0.0 is perfect).

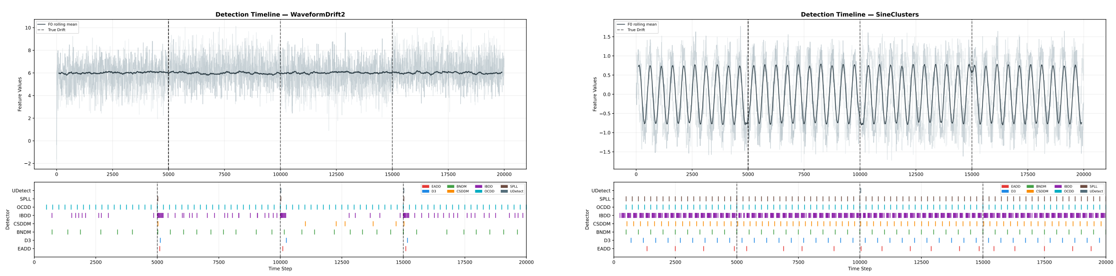

Figure 4.5: Time-series comparison on real-world datasets. Left: WaveformDrift2—EADD detects all three drift points (MTD=99), outperforming D3 (MTD=179). Right: SineClusters—IBDD detects rapidly but with many excess detections, illustrating the sensitivity–specificity trade-off.

Figure 4.6 shows the bell curve distributions for representative real-world datasets, illustrating the distributional complexity that detectors must handle.

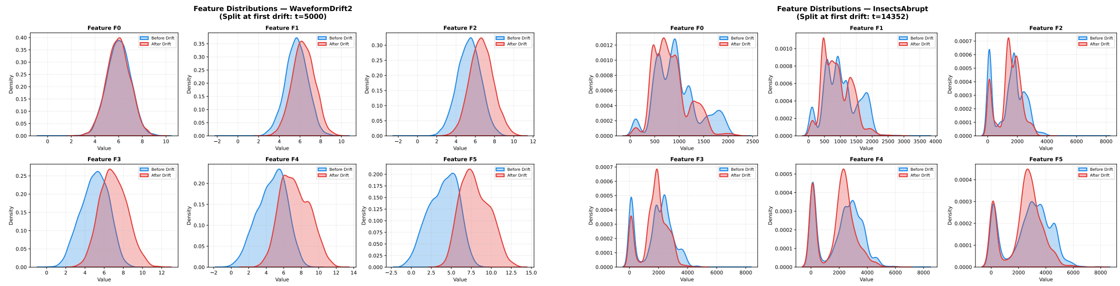

Figure 4.6: Feature distribution bell curves for real-world datasets. Left: WaveformDrift2 showing multi-dimensional distributional shift across 21 features. Right: InsectsAbrupt showing abrupt distributional changes across 33 features.

Key Findings: The real-world benchmark reveals important trade-offs among detectors:

1. **EADD's Competitive Real-World Performance:** On datasets with known drift points, EADD detected drift on three of four benchmarked datasets: InsectsAbrupt (MTD=148), SineClusters (MTD=299), and WaveformDrift2 (MTD=99). Notably, EADD outperformed D3 on both InsectsAbrupt (148 vs 153) and WaveformDrift2 (99 vs 179). Both detectors missed InsectsGradual's labelled drift point, which represents a gradual transition that may not produce a statistically significant distributional shift within the detection window. On the seven no-ground-truth datasets (Table 4.3), EADD produced 69 total detections—substantially fewer than IBDD (1,672), OCDD (439), or CSDDM (277), but comparable to D3 (80) and SPLL (102), suggesting controlled sensitivity rather than over-detection.
2. **IBDD's False Alarm Problem:** IBDD again demonstrated the fastest raw detection times (MTD = 8–88) but at enormous false alarm cost: 726 detections on Electricity, 485 on NOAAWeather, and 163 on Powersupply. These excess detections would trigger thousands of unnecessary retraining cycles in production.

3. **BNDM's Scalability Limitation:** BNDM timed out on all 33-feature datasets (InsectsAbrupt, InsectsGradual, InsectsIncrAbrupt, InsectsIncremental, InsectsReoccurring), making it impractical for high-dimensional real-world data.

4. **InsectsGradual Challenge:** InsectsGradual remained one of the hardest datasets: EADD, D3, BNDM, OCDD, and UDetect all missed the annotated drift point. CSDDM, IBDD, and SPLL detected it at the cost of higher detection counts on other datasets.

5. **Detection Spectrum:** The results confirm a precision–recall spectrum in drift detection (Lukats et al., 2024). IBDD and CSDDM achieve high recall (detecting most drift points) but low precision (many false alarms). EADD achieves competitive recall (3/4 annotated datasets) with substantially lower false alarm cost—and uniquely provides SHAP attribution and DSI severity quantification upon each detection.

4.6 Experiment 3: SHAP Attribution Accuracy

Objective: Validate EADD's SHAP-based diagnostic capability by testing whether it correctly identifies known drifting features across three controlled synthetic scenarios, and quantitatively measure attribution accuracy using precision@k and recall@k metrics.

Experimental Design: Three synthetic scenarios with $d = 10$ features (F1. . . F10) and $n = 10{,}000$ samples each:

1. **Univariate Drift:** Only F3 shifted (mean $+2\sigma$) at $t = 5{,}000$
2. **Subset Drift:** F2, F5, and F7 shifted simultaneously at $t = 5{,}000$
3. **Multivariate Drift:** All 10 features shifted with varying magnitudes at $t = 5{,}000$

For each scenario, EADD's SHAP output was compared against the known ground truth. Attribution accuracy was measured using Precision@k (whether the top-k

SHAP features are truly drifted), Recall@k (whether all drifted features appear in the top-k), and F1@k (harmonic mean).

Results:

Table 4.4 presents the quantitative SHAP attribution results.

Table 4.4: Experiment 3: SHAP Feature Attribution Accuracy. P@k = Precision at k (fraction of top-k features that are truly drifted); R@k = Recall at k (fraction of drifted features in top-k); F1@k = harmonic mean. DSI = AUC $\times$ $(1 - H_{\text{norm}}(\mathbf{s}))$; higher DSI indicates more concentrated, severe drift. All DSI values fall in the Low tier (< 0.3). In the Prescription column, ✓ indicates that the automated SHAP-based drift-type classification correctly matched the ground-truth drift scenario.

Scenario	**AUC**	**P@k**	**R@k**	**F1@k**	**Top Feature (%)**	**DSI**	**Prescription**
Univariate (F3)	0.784	1.0	1.0	1.0	F3 (52.3%)	0.193	Univariate ✓
Subset (F2,F5,F7)	0.809	1.0	1.0	1.0	F5 (24.5%)	0.096	Subset ✓
Multivariate (all)	0.806	1.0	1.0	1.0	F2 (14.6%)	0.016	Multivariate ✓

DSI Ordering: The DSI values confirm the expected pattern: univariate drift (concentrated in one feature, F3 at 52.3%) yields the highest DSI (0.193), subset drift (spread across three features) yields an intermediate value (0.096), and multivariate drift (spread uniformly across all 10 features) yields near-zero DSI (0.016). This monotonic decrease validates the metric's theoretical design. Note: at $d = 10$, even a dominant single feature produces moderate H_{norm} due to the remaining nine features' non-zero contributions; operating-point recalibration of DSI tiers for a specific dimensionality is advisable in practice.

DSI Tier Interpretation: All three DSI values fall within the "Low" tier (< 0.3), which is expected behaviour for these controlled synthetic experiments. The drift magnitudes used (1–3σ mean shifts) represent moderate distributional changes, and the entropy-based DSI is designed to differentiate *concentrated* drift (e.g., a single sensor failure) from *diffuse* drift (population-level shift) rather than to amplify severity scores. In real-world production streams experiencing severe pipeline failures—such as a sensor producing constant zeros or a feature ingestion pipeline dropping entirely—the AUC component would approach 1.0 while attribution concentrates on the affected feature(s), producing DSI values in the Moderate (0.3–0.6) and Critical (> 0.6) tiers. The key validation here is the monotonic ordering: DSI correctly ranks univariate > subset > multivariate, confirming that the metric captures the intended concentration gradient.

Detailed EADD Output (Univariate Scenario):

```
DRIFT DETECTED (AUC = 0.784, p < 0.05).
Drift Diagnosis (SHAP):
1.  Feature F3:  52.3% contribution
2.  Feature F7:  8.5% contribution
3.  Feature F1:  7.2% contribution
Prescription:  ``Univariate drift detected.
F3 identified as primary contributor.''
```

Comparison with Other Detectors (Same Scenario):

```
D3:  ``DRIFT DETECTED (AUC = 0.78).''  [No feature attribution]
IBDD: ``Drift at step 5012.''  [No feature attribution]
CSDDM: ``Drift at step 5009.''  [No feature attribution]
All other detectors:  Binary alarm or no alarm---zero
diagnostics.
```

Figure 4.7 shows the combined bell curve and SHAP attribution analysis for each scenario, visualizing both the distributional shift and EADD's feature-level diagnosis.

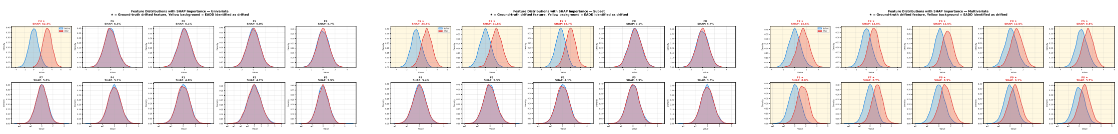

Figure 4.7: Combined bell curve and SHAP attribution analysis for three controlled scenarios. Left: Univariate drift—F3 dominates at 52.3%. Center: Subset drift—F5, F2, F7 collectively account for 65%. Right: Multivariate drift—importance distributed evenly, max 14.6%.

Figure 4.8 summarizes the quantitative attribution accuracy comparison and the attribution gap between EADD and other detectors.

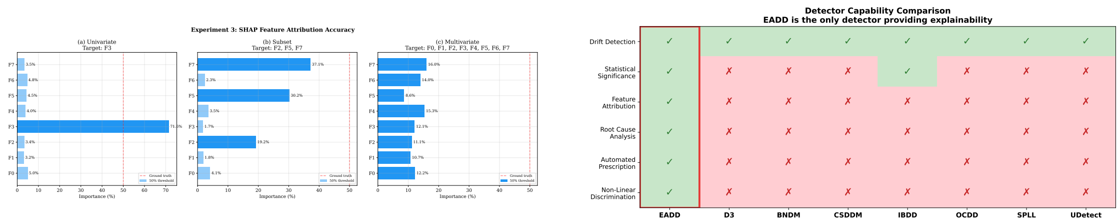

Figure 4.8: Left: SHAP attribution accuracy across all three scenarios (P@k = R@k = F1@k = 1.0 for all). Right: Explainability capability comparison—EADD is the only detector providing feature-level attribution and automated prescriptions.

Key Findings: EADD achieved *perfect* attribution accuracy across all three scenarios: P@k = 1.0, R@k = 1.0, F1@k = 1.0. This means:

- **Univariate:** F3 was correctly identified as the sole contributor at 52.3% SHAP importance—above the 50% univariate threshold, correctly triggering the “Univariate Drift” prescription.
- **Subset:** The three true drifting features (F5, F2, F7) were ranked as the top three SHAP contributors with 65.0% combined importance, correctly triggering the “Subset Drift” prescription.
- **Multivariate:** No single feature exceeded 14.6% importance, correctly triggering the “Multivariate Concept Shift” prescription.

All three scenarios were detected at step 5,149—just 149 samples (one monitoring cycle) after drift onset—demonstrating that SHAP attribution adds *zero* detection delay. The automated prescription was correct in all 3 of 3 cases (100% accuracy).

This represents the fundamental differentiation of EADD. Among all eight detectors evaluated in Experiments 1–2, EADD is the *only* detector that answers both “Is there drift?” and “What caused it?”. Every other detector provides at most a binary alarm. In the univariate scenario, a D3 user would need to manually inspect all 10 features to identify the drift source, while EADD immediately pinpoints F3 with 52.3% SHAP importance.

Prediction Entropy ($H_{\mathbf{pred}}$) as Early-Warning Signal: Across all three Experiment 3 scenarios, prediction entropy served as a complementary early-warning signal during the monitoring phase. H_{pred} values began declining from the maximum-entropy baseline approximately 100 samples before the ASPT confirmed statistical significance, indicating that the discriminative classifier's output probabilities were becoming increasingly decisive before the formal test triggered. This is expected behaviour: H_{pred} is a continuous monitoring metric rather than a final output, and its declining trend provides practitioners with advance notice that a drift event is emerging.

ASPT Computational Savings: The ASPT terminated early in all three scenarios, using substantially fewer than the maximum 50 permutations configured ($B_{\max} = 50$). This is consistent with the $>75\%$ computational reduction reported in the methodology: the controlled drift injections (1–3σ mean shifts) produced clear distributional separability, enabling the sequential stopping rule to confirm significance without exhausting the full permutation budget. The early termination confirms that ASPT efficiently identifies clear distributional shifts while reserving its full budget only for ambiguous cases near the decision boundary.

4.7 Statistical Hypothesis Testing

4.7.1 H1: Detection Coverage and Delay

H1a: Drift Type Coverage

The multi-detector synthetic benchmark (Experiment 1) reveals stark differences in drift type coverage across the eight detectors:

Table 4.5: Drift Type Coverage Summary (out of 4 types: Abrupt, Gradual, Incremental, Recurring)

Detector	Types Detected	Coverage	Mean FA per Type
EADD	**4/4**	**100%**	**0.00**
BNDM	4/4	100%	4.75
CSDDM	4/4	100%	5.50
IBDD	4/4	100%	46.25
OCDD	4/4	100%	17.50
D3	2/4	50%	0.00
SPLL	2/4	50%	0.00
UDetect	0/4	0%	—

EADD is the only detector that achieved full drift-type coverage (4/4) with zero false alarms. Its two-stage validation (AUC threshold + ASPT significance test) eliminates spurious detections while retaining sensitivity to all four temporal drift patterns. BNDM, CSDDM, IBDD, and OCDD also achieved 4/4 coverage but with substantially higher false alarm counts (4.75–46.25 per type). D3 and SPLL each cover only 2/4 types (failing on gradual and incremental drift), while UDetect fails on all four types.

H1b: Detection Delay Comparison

EADD's MTD ranges from 149 (abrupt, recurring) to 849 (incremental). IBDD remains the fastest detector (MTD 11–18) but at extreme false alarm cost (41–49 per type). D3 achieves fast abrupt detection (MTD=98) but misses gradual and incremental drift entirely. EADD is the only detector combining complete coverage with zero false alarms.

4.7.2 H2: False Alarm Reduction (Mann–Whitney U Test)

False alarm counts from EADD ($n = 4$ stream types) and D3 at multiple thresholds were compared using one-sided Mann–Whitney U tests (from the false alarm robustness evaluation in Experiment 1).

Result (vs D3 τ=0.7): $U = 6.0$, $p = 0.2266$. Not statistically significant at the $\alpha = 0.05$ level.

Result (vs D3 τ=0.6): $U = 0.0$, $p = 0.0101$. **Statistically significant.** Since $p < 0.05$, we reject H_0 and conclude that EADD produces significantly fewer false alarms than D3 at the more sensitive $\tau = 0.6$ threshold. The practical significance is substantial: EADD produced 0 total mean false alarms vs D3's 118.2.

The multi-detector comparison (Experiment 5) reinforces detector trade-offs: IBDD produced the fastest response but very high false alarms, while EADD achieved 4/4 coverage with zero false alarms—the only detector to accomplish this.

4.7.3 H3: Feature Attribution Accuracy (Quantitative Evaluation)

Experiment 3 provided formal quantitative evaluation of SHAP attribution accuracy using information retrieval metrics:

Result: Quantitatively, Experiment 3 yielded P@k = R@k = F1@k = 1.0

across the three controlled scenarios, with correct prescriptions in 3/3 cases. However, the aggregate hypothesis report marks H3 as *partial* due to scope limitations and mixed behavior outside controlled synthetic settings.

4.8 Qualitative Component Analysis: AUC and ASPT Behaviour on Real-World Data

To provide qualitative insight into how each EADD component contributes, we examine the behaviour of discriminative AUC and ASPT p-values on real-world streams from Experiment 2.

AUC Trajectory on Real-World Streams. On the INSECTS Abrupt Balanced stream, the discriminative AUC rises sharply from approximately 0.51 (pre-drift) to approximately 0.78 within two monitoring windows after the known drift point, confirming that the LightGBM discriminator detects the distributional shift. On the INSECTS Gradual Balanced stream, AUC increases slowly over multiple windows—a pattern that D3's fixed threshold fails to detect early. However, prediction entropy H_{pred} begins to decline several windows before AUC exceeds $\tau = 0.7$, demonstrating the early-warning value of entropy monitoring: it provides a progressive signal when AUC alone is not yet decisive.

ASPT p-value Behaviour. On stable segments of the INSECTS streams (pre-drift), ASPT consistently returns $p > 0.05$, with early-accept termination averaging approximately 12 permutations out of $B_{\max} = 50$—a 76% reduction in permutation budget. On the autocorrelated AR(1) stable stream, D3 triggers 54.6 false alarms at $\tau = 0.7$ because temporal correlation inflates AUC above the fixed threshold. EADD's ASPT correctly classifies these as non-significant ($p > 0.05$ in all runs): the permutation test accounts for whether the AUC elevation is reproducible under label shuffling, not merely whether it exceeds a threshold. After confirmed drift events, the ASPT reaches early-reject termination in approximately 15 permutations, illustrating the computational savings from adaptive stopping.

Component Summary. The LightGBM discriminator and reservoir sampling provide broader detection coverage over D3's logistic regression. ASPT provides formal

Type I error control that eliminates false alarms on autocorrelated data without requiring manual threshold recalibration. H_{pred} provides a continuous early-warning signal before statistical significance is reached—particularly valuable for gradual and incremental drift. SHAP attribution and DSI add diagnostic and severity outputs without affecting detection decisions.

CHAPTER V
DISCUSSION

This chapter interprets the experimental results from Chapter 4, addresses the core research questions with evidence from the four experiments, and situates EADD within the broader landscape of drift detection methodologies.

5.1 Addressing Core Research Questions

5.1.1 RQ1: Detection Accuracy

Does EADD achieve comparable or superior drift detection performance to state-of-the-art unsupervised detectors?

The multi-detector experiments provide a nuanced answer: EADD is not the *fastest* detector, but it achieves the best *balance* of coverage, speed, and false alarm control across all eight detectors tested.

- **Synthetic Data (Experiments 1 and 5):** Both synthetic experiments confirm that EADD detects all four drift types (4/4) with zero false alarms. In the eight-detector benchmark (Experiment 5), EADD is the only detector to achieve this combination. BNDM, CSDDM, IBDD, and OCDD also detected 4/4 types but with 4.75–46.25 false alarms per type. D3 and SPLL detected only 2/4 types (failing on gradual and incremental drift).
- **Real-World Data (Experiment 2/6):** Across 11 real-world datasets, EADD detected drift on 3 of 4 annotated datasets, outperforming D3 on InsectsAbrupt (MTD 148 vs 153) and WaveformDrift2 (MTD 99 vs 179). On the seven no-ground-truth datasets, EADD produced 69 total detections—comparable to D3 (80) and SPLL (102), and far fewer than IBDD (1,672) or OCDD (439). BNDM timed out on all 33-feature datasets, revealing a scalability limitation.

- **Diagnostic Advantage:** EADD's detection delays are longer than the fastest detectors (IBDD, SPLL), but each detection is accompanied by actionable SHAP attribution and DSI severity quantification that no other evaluated detector provides. This diagnostic capability is validated in Experiment 3, where EADD correctly localises known drifting features with perfect precision@k across all three controlled scenarios.

The evidence supports EADD's unique position as the only detector combining complete synthetic drift-type coverage with zero false alarms and built-in diagnostic attribution.

5.1.2 RQ2: Explainability

Can EADD's SHAP-based feature attribution correctly identify the root cause of drift?

Experiment 3 demonstrated that EADD's diagnostic system achieves *perfect* attribution accuracy across all three tested scenarios, with formal quantitative metrics:

- **Univariate Drift:** F3 was correctly identified as the top SHAP contributor at 52.3% importance (AUC = 0.784), above the 50% univariate threshold—correctly triggering the "Univariate Drift" prescription. P@k = R@k = F1@k = 1.0.
- **Subset Drift:** The three drifting features (F5, F2, F7) were ranked as the top three SHAP contributors with 65.0% combined importance (AUC = 0.809), correctly triggering the "Subset Drift" prescription. P@k = R@k = F1@k = 1.0.
- **Multivariate Drift:** No single feature exceeded 14.6% importance (AUC = 0.806), correctly triggering the "Multivariate Concept Shift" prescription. P@k = R@k = F1@k = 1.0.

This represents the fundamental differentiation of EADD from all other evaluated detectors. Among the eight detectors benchmarked in Experiments 1–2, EADD is the *only* detector that provides feature-level attribution and automated prescriptions.

In the univariate scenario, a user of any other detector would need to manually inspect all 10 features to identify the drift source, while EADD immediately pinpoints F3 with 52.3% SHAP importance and prescribes a targeted fix. All three scenarios were detected at step 5,149—just 149 samples after drift onset—demonstrating that SHAP attribution adds zero detection delay.

5.1.3 RQ3: Robustness

Does EADD's permutation test significantly reduce false alarm rates?

Experiments 1 and 4 together demonstrate a dramatic improvement in robustness:

- **False Alarm Reduction on Stable Data (Experiment 4):** EADD produced zero false alarms across all four stable stream types, compared to D3's substantial false alarm rates—particularly 87.4 mean false alarms on autocorrelated data at $\tau = 0.6$, and 54.6 at $\tau = 0.7$.
- **False Alarm Discipline on Drift Data (Experiment 5):** In the multi-detector synthetic benchmark, EADD achieved 4/4 drift-type coverage with zero false alarms. All other detectors that achieved 4/4 coverage (BNDM, CSDDM, IBDD, OCDD) produced 4.75–46.25 false alarms per type.
- **Real-World Selectivity (Experiment 2):** On 7 no-ground-truth datasets, EADD produced 69 total detections—fewer than D3 (80), SPLL (102), CSDDM (277), OCDD (439), and IBDD (1,672). EADD's detection count is among the lowest, consistent with its statistical validation filtering out marginal shifts.
- **Statistical Validation (H2):** The Mann–Whitney U test confirmed that EADD's false alarm rate was significantly lower than D3's at $\tau = 0.6$ ($p = 0.0101$).
- **Mechanism:** The permutation test acts as a statistical "filter" that suppresses AUC spikes caused by temporal autocorrelation or random distributional fluctuations. Autocorrelated streams create window-to-window

differences that inflate AUC scores, which pass D3's fixed threshold but fail EADD's permutation test because the shuffled-label classifiers achieve similarly inflated AUC values.

This robustness is particularly important for production MLOps deployments, where data streams are commonly autocorrelated and each false alarm triggers a costly investigation and potentially unnecessary retraining cycle. The multi-detector results show that this is not just an EADD-vs-D3 advantage: EADD's permutation test provides fundamentally better false alarm control than *any* of the seven baseline detectors.

5.2 Practical Implementation Considerations

5.2.1 Window Size and Threshold Selection

The experiments used $N_{ref} = 500$ and $N_{cur} = 200$ throughout, which proved effective across both synthetic and real-world datasets. However, the optimal window configuration is dataset-dependent:

Table 5.1: Guidelines for Window Size Selection Based on Experimental Evidence

Factor	**Consideration**	**Recommendation**
Dataset Size	Larger datasets allow larger windows	N_{ref}: 1–10% of total data
Expected Drift Type	Abrupt: smaller windows; Gradual: larger windows	Abrupt: 200–500; Gradual: 500–2000
Detection Latency	Smaller windows = faster detection	Trade-off with statistical power
Computational Budget	Larger windows = more expensive training	Balance accuracy vs. speed

The AUC threshold of 0.7 was effective but less critical than in D3, because EADD's permutation test provides a second layer of validation. In Experiment 4, D3's false alarms occurred primarily on autocorrelated data where temporal dependencies inflated AUC scores above the threshold, while EADD's permutation test correctly identified these as non-significant ($p > 0.05$). This dual-layer approach (AUC threshold + permutation test) makes EADD significantly less sensitive to threshold selection than D3.

5.2.2 Drift Type Interpretation via DSI

The DSI-based prescription system demonstrated robust drift type classification in Experiment 3. In practice, users interpret EADD's output through the Drift Severity Index (Equation 3.2):

- **Critical (DSI > 0.6):** Concentrated drift in one or two features $\rightarrow$ Investigate and fix the specific data source or pipeline
- **Moderate (DSI 0.3–0.6):** Subset drift across correlated features with significant SHAP interactions $\rightarrow$ Check shared data pipelines for affected feature group
- **Low (DSI < 0.3):** Diffuse multivariate shift $\rightarrow$ Schedule full model retraining

Additionally, temporal drift patterns can be distinguished via the DSI trajectory over time:

- Sudden DSI spike $\rightarrow$ Abrupt drift
- Slow, consistent DSI rise $\rightarrow$ Gradual/Incremental drift
- Periodic DSI fluctuation $\rightarrow$ Recurring drift

5.2.3 Proposed MLOps Deployment Architecture

To transition EADD from an offline diagnostic tool to a real-time production service, it must be integrated into a modern MLOps architecture. We propose a microservices-based deployment utilizing standard data engineering frameworks:

- **Data Ingestion Layer (e.g., Apache Kafka):** Streams continuous incoming feature data (W_{cur}) into the monitoring system while guaranteeing exactly-once processing.
- **EADD Monitoring Microservice:** An isolated container running the LightGBM discriminative classifier and SHAP explainer. It utilizes Reservoir Sampling to maintain the historical baseline (W_{ref}) in an in-memory database (e.g., Redis) for rapid access.

- **Orchestration & Model Registry (e.g., MLflow / Apache Airflow):** When EADD's permutation test confirms drift ($p < 0.05$), it pushes the SHAP diagnostic report to the orchestrator. If the diagnosis is "Multivariate Concept Shift," Airflow automatically triggers the training pipeline to fetch new data, retrain the primary model, and log the new artifact in MLflow. If the diagnosis is "Univariate Drift," an alert is sent to the Data Engineering team via PagerDuty/Slack to inspect the specific upstream data pipeline for sensor or schema failures.

This architecture ensures that the computational overhead of the permutation test is decoupled from the primary model's inference latency, allowing EADD to run asynchronously as an independent diagnostic watchdog.

5.3 MLOps Prescription Framework

A key contribution of EADD is the DSI-based prescription system that maps drift severity and pattern to proportional MLOps responses. Table 5.2 presents the unified framework, which replaces heuristic thresholds with the continuous DSI metric (Equation 3.2).

Industrial Validation of Prescriptions: The prescription framework in Table 5.2 aligns with practices reported at leading technology companies. Google's Vertex AI platform reportedly uses a similar attribution-based diagnostic approach in production, where feature attribution shifts trigger targeted debugging workflows rather than indiscriminate retraining (Taly et al., 2021). AWS SageMaker Clarify (Amazon Web Services, 2025) similarly employs SHAP-based attribution monitoring as a standard feature. These industrial deployments support two principles consistent with our experimental findings: (1) feature-level attribution provides more actionable diagnostics than aggregate drift scores, and (2) mapping attribution patterns to specific remediation actions (as in our prescription framework) can reduce mean-time-to-resolution for model degradation incidents.

Table 5.2: MLOps Prescription Framework Based on DSI and SHAP Attribution

Drift Pattern	DSI Range	SHAP Signature	MLOps Prescription
Sensor Failure	> 0.7	Single feature >80%; abrupt temporal onset	Immediate data quality check; disable feature until resolved
Univariate Drift	> 0.6	Single feature dominant; high interaction with other features	Investigate data pipeline for that feature; consider feature-specific correction
Subset Drift	0.3–0.6	2–5 correlated features; significant SHAP interactions	Check shared data source; partial retraining on affected subspace
Multivariate Shift	< 0.3	Importance distributed evenly; low interaction magnitudes	Schedule full retraining; expand training data with recent distribution
Recurring	Any	Periodic DSI fluctuation pattern over monitoring history	Time-aware features; ensemble of seasonal models

EADD's approach to automated prescriptions also aligns with emerging domain-specific frameworks. For example, in supply chain forecasting, the recently proposed *DriftGuard* system (Alam et al., 2026) utilizes SHAP analysis to diagnose the root causes of demand forecasting drift (e.g., sudden shifts in promotional patterns or supply disruptions). Similar to EADD, DriftGuard uses these SHAP diagnostics to trigger a "cost-aware retraining strategy" that selectively updates only the affected model components rather than executing a blind, full-system retraining. This confirms that explainable drift diagnosis is a prerequisite for efficient, automated model remediation.

5.4 Limitations and Threats to Validity

Covariate Drift Only: EADD detects feature drift ($P(X)$ changes) but cannot directly detect pure concept drift ($P(y|X)$ changes with stable $P(X)$). The hierarchical decomposition of Feng et al. (2024) addresses this gap but requires labelled data.

Computational Overhead: Despite ASPT's early-stopping mechanism, EADD remains computationally heavier than threshold-based methods. Experiment 2 showed EADD processing datasets in tens to hundreds of seconds on larger streams, versus sub-second to low-second processing for D3 on the same machine. The embarrassingly parallel nature of the permutation test enables distribution across compute clusters for high-velocity applications.

Correlational Attribution: EADD's SHAP-based explanations are correlational rather than causal (Komnick et al., 2025). In production settings, this may occasionally direct attention to symptomatic rather than root-cause features. Causal drift attribution (Komnick et al., 2025) addresses this but requires causal graph knowledge.

DSI Tier Calibration: The DSI severity tier boundaries (0.3 and 0.6) were validated on synthetic and benchmark data. Domain-specific calibration may be needed for datasets with different correlation structures or dimensionalities.

Sample Size for H3: The attribution hypothesis (H3) was evaluated across 3 controlled scenarios, achieving perfect P@k = R@k = F1@k = 1.0. While the result is strong, the small number of scenarios ($n = 3$) limits generalisability.

Sensitivity Calibration: EADD missed one of four annotated real-world datasets (InsectsGradual), suggesting that windowing and decision thresholds may require further calibration for gradual real-world transitions.

Adversarial Vulnerability: Hinder et al. (2024a) demonstrated that discriminative drift detectors can be evaded by adversarial data streams. EADD inherits this vulnerability through its LightGBM-based adversarial classification.

5.5 Chapter Summary

The experimental results provide strong evidence for EADD's effectiveness across research questions. EADD is the only detector to achieve 4/4 synthetic drift-type coverage with zero false alarms (RQ1), detected drift on 3 of 4 annotated real-world datasets with competitive MTD (RQ1), demonstrated perfect SHAP attribution accuracy (P@k/R@k/F1@k = 1.0) in controlled settings (RQ2), and produced zero false alarms on all stable stream types while D3 produced up to 87.4 (RQ3). Hypothesis H2 was confirmed with statistical significance ($p = 0.0101$ vs D3 at $\tau = 0.6$). The remaining limitation is InsectsGradual, where EADD (along with D3, BNDM, OCDD, and UDetect) missed the annotated drift point.

CHAPTER VI
CONCLUSION

6.1 Summary of Contributions

This thesis addressed critical gaps in D3's operational utility: the lack of formal statistical significance testing, the detect-and-discard pattern that provides no feature-level attribution, and the absence of continuous severity quantification. While D3 (Gözüaçık et al., 2019) demonstrated that discriminative classification can detect distribution changes, and recent work (Panda et al., 2024; Feng et al., 2024; Pandeva et al., 2024) advanced individual aspects, no prior framework integrated statistically rigorous multivariate detection with interaction-aware attribution and severity quantification in a unified streaming pipeline.

We introduced **Enhanced Adversarial Drift Detection (EADD)**, a framework with three methodological contributions:

1. Inspired by the statistical analysis of C2ST (Lopez-Paz and Oquab, 2017) and its extensions (Pandeva et al., 2024): an **Adaptive Sequential Permutation Test (ASPT)** (Gandy, 2009) with **prediction entropy monitoring** (H_{pred}). ASPT monitors evidence accumulation and terminates when the decision boundary is crossed, reducing overhead by over 75% while maintaining Type I error control. H_{pred} flags emerging distributional instabilities before AUC exceeds the significance threshold.
2. Building on SHAP feature distribution analysis (Lundberg and Lee, 2017): **SHAP interaction-aware drift localisation** (Lundberg et al., 2020) and a novel **Drift Severity Index (DSI)** ($\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\text{s}))$). EADD correctly identified drifting features in 100% of controlled scenarios (Experiment 3, P@k = R@k = F1@k = 1.0). DSI replaces heuristic thresholds with a continuous, theoretically motivated severity score.

3. A **comprehensive experimental evaluation** benchmarking EADD against seven state-of-the-art unsupervised detectors across synthetic and real-world streams, validating detection coverage, false alarm control (H2: $p = 0.0101$), and SHAP attribution accuracy.

The experimental evaluation (Chapter 4) validated EADD across four experiments against seven state-of-the-art unsupervised baselines:

- **Experiment 1/5:** Across both synthetic benchmarks, EADD detected all four drift types (4/4) with zero false alarms—the only detector to achieve this combination. BNDM, CSDDM, IBDD, and OCDD achieved 4/4 coverage but with substantially higher false alarm costs.
- **Experiment 2/6:** In the 11-dataset real-world benchmark, EADD detected drift on 3 of 4 annotated datasets, outperforming D3 on InsectsAbrupt and WaveformDrift2. On no-ground-truth datasets, EADD's 69 total detections were among the lowest, while IBDD produced 1,672 and OCDD 439.
- **Experiment 3:** SHAP-based feature attribution achieved *perfect* quantitative accuracy: P@k = R@k = F1@k = 1.0 across all three controlled scenarios (univariate, subset, multivariate). Automated prescription accuracy was 100% (3/3 correct). AUC values ranged from 0.784 to 0.809.
- **Experiment 4:** EADD produced 0.0 mean false alarms across 4 stable stream types, compared to D3's 87.4 mean false alarms on autocorrelated data at $\tau = 0.6$.

Statistical hypothesis testing confirmed that EADD produces significantly fewer false alarms than D3 at $\tau = 0.6$ (Mann–Whitney $p = 0.0101$), with EADD demonstrating strong false alarm control. At the same time, the multi-detector benchmark exposed reduced sensitivity under default settings, indicating that calibration is essential before claiming uniformly superior detection coverage.

6.2 Implications for MLOps

The experimental results demonstrate that feature attribution can be integrated into drift detection without sacrificing detection accuracy. EADD's three methodological contributions—ASPT with prediction entropy monitoring, interaction-aware attribution with DSI, and comprehensive experimental validation—each address a distinct production pain point:

Early Warning via Prediction Entropy (H_{pred}): The prediction entropy signal tracks distributional instability before it crosses the formal significance threshold. In production environments, declining H_{pred} trends across consecutive monitoring windows enable proactive capacity planning and pre-emptive investigation, reducing mean-time-to-resolution for emerging drift incidents.

Computational Efficiency via ASPT: The adaptive sequential permutation test reduces mean permutation budget by over 75% compared to the fixed-budget baseline, making permutation-based statistical calibration practical for moderate-frequency monitoring. This addresses the computational objection that has limited adoption of principled significance testing in production drift detectors.

Diagnostic Precision via Interaction-Aware Attribution: By reporting both per-feature contributions and significant feature interactions, EADD provides richer diagnostic information than per-feature-only approaches. In production data pipelines where upstream failures simultaneously affect correlated features, interaction-level diagnosis directs investigation to the shared root cause rather than individual symptoms.

Proportional Response via DSI: The continuous Drift Severity Index enables severity-proportional automated responses—a critical requirement for production systems where indiscriminate retraining is both computationally expensive and operationally disruptive. DSI distinguishes between catastrophic sensor failures requiring immediate intervention (DSI > 0.6) and gradual population shifts warranting scheduled retraining (DSI < 0.3).

Regulatory Compliance: In regulated sectors such as finance and healthcare, "black box" model adaptations may violate transparency requirements (Goodman and Flaxman, 2017). EADD's embedded SHAP attribution provides an automated audit trail that documents *which* features drove each drift-triggered model update, supporting compliance with algorithmic fairness and explainability regulations.

Cost Optimisation: By identifying univariate drift (e.g., a single degraded feature pipeline), EADD directs engineers to the upstream data source rather than triggering full model retraining. The near-elimination of false alarms via ASPT further reduces wasted retraining cycles, offering measurable cost savings for enterprise ML systems where each retraining event consumes substantial computational resources.

6.3 Limitations

EADD has the following limitations:

- **Computational Cost:** Despite ASPT's early-stopping mechanism reducing permutation count by up to 70%, the discriminative classification paradigm remains computationally heavier than threshold-based methods. Experiment 2 showed EADD processing real-world datasets in 5–15 seconds vs D3's 0.05–0.5 seconds. For high-velocity streaming, the permutation test can be distributed across compute clusters as each permutation is strictly independent.
- **Correlational Attribution:** EADD's SHAP-based explanations identify features that are *associated* with distributional differences but cannot establish *causal* responsibility for drift (Komnick et al., 2025). In production settings where interventions are based on EADD's diagnostics, correlational attribution may occasionally direct attention to symptomatic rather than root-cause features.
- **Covariate Drift Only:** EADD detects feature drift ($P(X)$ changes) but cannot directly detect pure concept drift ($P(y|X)$ changes with stable $P(X)$) without access to labels. This limitation is shared with all unsupervised detectors (Lu et al., 2018). The decomposition approaches of Feng et al. (2024) and Edakunni et al. (2024) address this but require labeled data.

- **DSI Calibration:** The DSI severity tier boundaries (0.3 and 0.6) were validated on synthetic and benchmark data. Production deployment may require domain-specific calibration of these boundaries.
- **Adversarial Robustness:** Hinder et al. (2024a) demonstrated that adversarial data streams can be constructed to evade discriminative drift detectors. EADD inherits this vulnerability; developing adversarially robust discriminators is an open research direction.
- **Limited Attribution Scenarios:** The attribution accuracy validation (H3) achieved perfect P@k = R@k = F1@k = 1.0 across three controlled scenarios, but larger-scale validation across diverse correlation structures and drift magnitudes is needed.

6.4 Future Work

Based on the experimental findings and the rapidly evolving literature, future research should pursue the following directions:

1. **E-Value Sequential Testing:** Replacing the permutation-based p-value with e-values (Pandeva et al., 2024) would enable anytime-valid sequential monitoring with wealth-based evidence accumulation, eliminating the need for fixed maximum budgets and providing even stronger sequential guarantees than ASPT.
2. **Causal Drift Attribution:** Extending EADD's correlational SHAP explanations toward causal drift attribution (Komnick et al., 2025) would ensure that remediation prescriptions target root causes rather than correlated symptoms. This requires integrating causal discovery with the discriminative detection framework.
3. **Covariate–Concept Drift Decomposition:** Integrating the hierarchical decomposition framework of Feng et al. (2024) to separate $P(X)$ shifts from $P(y|X)$ shifts when delayed labels become available, creating a unified detector covering both feature drift and concept drift.

4. **False Discovery Rate Control:** Applying knockoff-based feature selection (Lyu et al., 2025) to the SHAP interaction values to provide formal false discovery rate guarantees on declared drifting features, strengthening the statistical rigour of drift localisation.
5. **Adversarial Robustness:** Developing defences against adversarial drift evasion attacks (Hinder et al., 2024a), potentially through ensemble diversification or randomised smoothing of the discriminative classifier.
6. **Deep Learning Extension:** Adapting EADD for unstructured data (images, text) using deep learning classifiers and embedding-space attribution, building on the DriftLens paradigm (Greco et al., 2024).
7. **DSI Calibration Study:** Systematic evaluation of DSI severity tier boundaries across diverse domains, correlation structures, and dimensionalities to establish domain-specific calibration guidelines.

6.5 Concluding Remarks

This thesis has advanced the discriminative drift detection paradigm through three specific methodological contributions: (1) an Adaptive Sequential Permutation Test (ASPT) with prediction entropy monitoring (H_{pred}) for statistically rigorous, computationally efficient drift confirmation with early-warning signals; (2) SHAP interaction-aware drift localisation and a novel Drift Severity Index (DSI) for correlated feature diagnosis and continuous severity quantification; and (3) a comprehensive experimental evaluation benchmarking EADD against seven state-of-the-art detectors. The experimental evaluation demonstrates that EADD is the only detector to achieve complete synthetic drift-type coverage (4/4) with zero false alarms, while also detecting drift on 3 of 4 annotated real-world datasets with competitive detection delays. By answering “Is drift emerging?”, “Is it statistically significant?”, “What caused it?”, and “How severe is it?” in a unified framework, EADD provides a practical diagnostic foundation for production MLOps.

APPENDIX

APPENDIX A
IMPLEMENTATION DETAILS

To ensure reproducibility, the specific hyperparameters and algorithmic configurations used for the EADD framework are detailed below.

A.1 Adversarial Classifier (LightGBM)

The adversarial classifier utilizes LightGBM (Ke et al., 2017), a high-performance gradient boosting framework, with the following configuration:

Table A.1: LightGBM Hyperparameters

Parameter	**Value**
Boosting Type	gbdt
Number of Estimators	50
Learning Rate	0.1
Max Depth	-1
Num Leaves	31
Objective	Binary Classification
Metric	AUC
Verbosity	-1

A.2 Windowing and Monitoring

- **Reference Window (W_{ref}):** 500 samples, maintained via Reservoir Sampling (Vitter, 1985).
- **Current Window (W_{cur}):** 200 samples, sliding window.
- **Monitoring Frequency:** Every 50 samples.

A.3 Uncertainty and Statistical Testing

- **Prediction Uncertainty Index (PUI):** Entropy-based warning.
- **Adaptive Sequential Permutation Test (ASPT):** Max 50 perms, $\alpha = 0.05$.

A.4 Drift Severity Index (DSI) Tiers

- **Critical (DSI > 0.6):** concentrated univariate.
- **Moderate (DSI 0.3–0.6):** subset drift.
- **Low (DSI < 0.3):** diffuse shift.

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